

Table 2-3. Land Use

Land-Use Type	Upper Shenango River Subwatershed		Middle Shenango River Subwatershed		Lower Shenango River Subwatershed		Neshannock Creek/Big Run Subwatershed		Shenango River Watershed	
	Square Miles	% of Land Area	Square Miles	% of Land Area	Square Miles	% of Land Area	Square Miles	% of Land Area	Square Miles	% of Land Area
Agriculture	55.01	35.51	213.94	49.16	80.27	38.53	149.89	55.36	499.11	46.68
Pasture/Hay	37.17	23.99	152.73	35.10	62.08	29.80	135.09	49.90	387.07	36.20
Row Crops	17.84	11.51	61.21	14.07	17.22	8.27	14.80	5.47	111.07	10.39
Urban/Recreational Grasses	0.00	0.00	0.00	0.00	0.97	0.46	0.00	0.00	0.97	0.09
Forest	59.65	38.50	175.96	40.44	91.69	44.01	98.32	36.31	425.62	39.81
Deciduous Forest	51.78	33.42	160.87	36.97	83.89	40.27	88.63	32.73	385.17	36.03
Evergreen Forest	3.53	2.28	4.68	1.08	2.33	1.12	1.73	0.64	12.27	1.15
Mixed Forest	4.34	2.80	10.41	2.39	5.47	2.63	7.96	2.94	28.18	2.64
Wetlands	8.93	5.76	24.26	5.57	8.74	4.19	12.08	4.46	54.01	5.05
Woody Wetlands	8.13	5.25	21.68	4.98	7.46	3.58	9.35	3.45	46.62	4.36
Emergent Wetlands	0.80	0.52	2.58	0.59	1.28	0.61	2.73	1.01	7.39	0.69
Development	2.98	1.92	10.92	2.51	25.68	12.32	8.83	3.26	48.40	4.53
Low Density Urban	2.68	1.73	8.89	2.04	16.44	7.89	6.27	2.32	34.28	3.21
High Density Urban	0.07	0.24	0.32	3.32	1.84	127.94	0.15	32.42	2.37	6.01
Commercial/Industrial	0.22	0.15	1.71	0.39	7.40	3.55	2.41	0.89	11.74	1.10
Water	28.05	18.11	9.49	2.18	1.44	0.69	0.46	0.17	39.43	3.69
Open Water	28.05	18.11	9.49	2.18	1.44	0.69	0.46	0.17	39.43	3.69
Barren	0.32	0.21	0.59	0.14	0.52	0.25	1.18	0.43	2.61	0.24
Mining	0.00	0.00	0.57	0.13	0.09	0.04	1.08	0.40	1.73	0.16
Transitional	0.32	0.21	0.02	0.01	0.43	0.21	0.10	0.04	0.87	0.08

Active Mines

Mining operations must have an active permit for the site they are mining to be considered an active mine. Even though the permits may be active, mining may not be physically occurring in the permitted areas. There are various stages to active permits including: not started, active, treatment, reclamation, and forfeited. Permits are generally issued for an area larger than the company is planning to mine to support the movement of material and equipment.

Industry and coal mining account for less than one percent of the land use within the Shenango River watershed. Currently there are 51 active or pending permits. Industrial mineral mining is the dominant mining industry in the watershed. Mining is occurring throughout the entire watershed; however, the

majority of mining in the watershed is occurring in the Middle Shenango River and Lower Shenango River subwatersheds.

The only active coal-mining permit in the Shenango River watershed is located in the Neshannock Creek/Big Run subwatershed. The mine is operated by Amerikohl Mining Company and located in Washington Township, Lawrence County.

A listing of active and pending permits and their status for the Shenango River watershed is available in Appendix D.



*Steel plant along the Shenango River
in the Lower Shenango River
subwatershed*

Agriculture

Trends show that the numbers of family farms are declining while the sizes of farms are increasing. The declining economy and the pressure to meet increased government regulations are causing financial problems for many small farms. Support for agriculture is needed since it is the primary land use in the watershed.

Industrial/Commercial

Approximately 12 square miles, one percent of the total land use of the Shenango River watershed, are used for commercial and industrial purposes. The majority of the industry in the Shenango River watershed, 63 percent, is located in the Lower Shenango River subwatershed near Sharon and Farrell. Another 20 percent is located in the Neshannock Creek/Big Run subwatershed.

Waste Disposal

Landfills

According to the Pennsylvania Department of Environmental Protection (PA DEP), there are currently no active landfills located within the Shenango River watershed. Records from the PA DEP indicate there are five closed landfills within the watershed. They include Henry's Disposal Service Site 1, Henry's Disposal Service Site 2, Kwiatkowski Sanitary Landfill, Carlson Mining Sanitary Landfill, and River Road Landfill. The waste at the Henry's Disposal Service sites is assumed to be from Henry's Disposal Service. It has not been confirmed. The sites are located in Linesville, Crawford County. Kwiatkowski Sanitary Landfill is located in Hickory Township, Lawrence County, and the Carlson Mining Sanitary Landfill is located in Slippery Rock Township, Lawrence County. The River Road Landfill is an abandoned landfill in Mercer County. This site was listed on the Superfund list, but was removed in January 2004 after remediation efforts were completed.

Recycling

Recycling involves the separation and collection of materials such as paper, aluminum, and steel for processing and remanufacturing into new products. Recycling helps conserve natural resources, reduce pollution, saves space in landfills, creates jobs, and saves money.

In July 1988, Act 101 was passed mandating recycling in 415 Pennsylvania communities with populations greater than 5,000. Since Act 101 was enacted, more than 450 communities have voluntarily developed curbside recycling programs, and more than 750 communities are serviced by drop-off centers

for recyclables. Within the Pennsylvania portion of the watershed, solid waste authorities in each county coordinate recycling programs.

Ownership

The majority of the watershed, 93.5 percent, is privately owned. These private holdings include residential areas, farmland, forested areas, commercial and industrial properties, and private camps. Public properties within the watershed include Pymatuning State Park, Shenango River Lake, State Game Lands, schools, community parks, and municipal properties.

Pymatuning State Park contains 21,122 acres in Pennsylvania and Ohio, making up the majority of the Upper Shenango River watershed. Shenango River Lake accounts for 15,071 acres of the Middle Shenango River watershed. State Game Lands make up 7,858 acres of land available for public use.

Critical Areas

Critical areas are areas having constraints that limit development and various other activities. Critical natural areas are those areas containing rare, threatened, or endangered species, natural communities of special concern, or significant ecological and geological landscapes worthy of protection. Steep slopes, ridgetops, floodplains, streambanks, and wetlands are examples of critical natural areas. Figure 2-11 identifies environmentally sensitive areas.

Landslides

Landslides occur throughout Pennsylvania and are heavily concentrated in southwestern Pennsylvania. Most landslides occur in areas with steep slopes where loose colluvial soils exist. However, landslides can also occur in areas containing clay soils, because the characteristics of clay soils vary significantly with water content. Gravity eventually forces this rock and debris down the slope in a gradual or sudden, flashy manner. Landslides can also occur as a slump, where a block of weathered rock or soil slides outward because of the force from the weight rotation of weathered rock or soil above it. Typically, landslides occur along road cuts with unstable bank conditions. Other factors, such as stream erosion, earth moving activities, soil characteristics, weakened or fractured rock, mining debris, and weather, can determine the occurrence of a landslide.



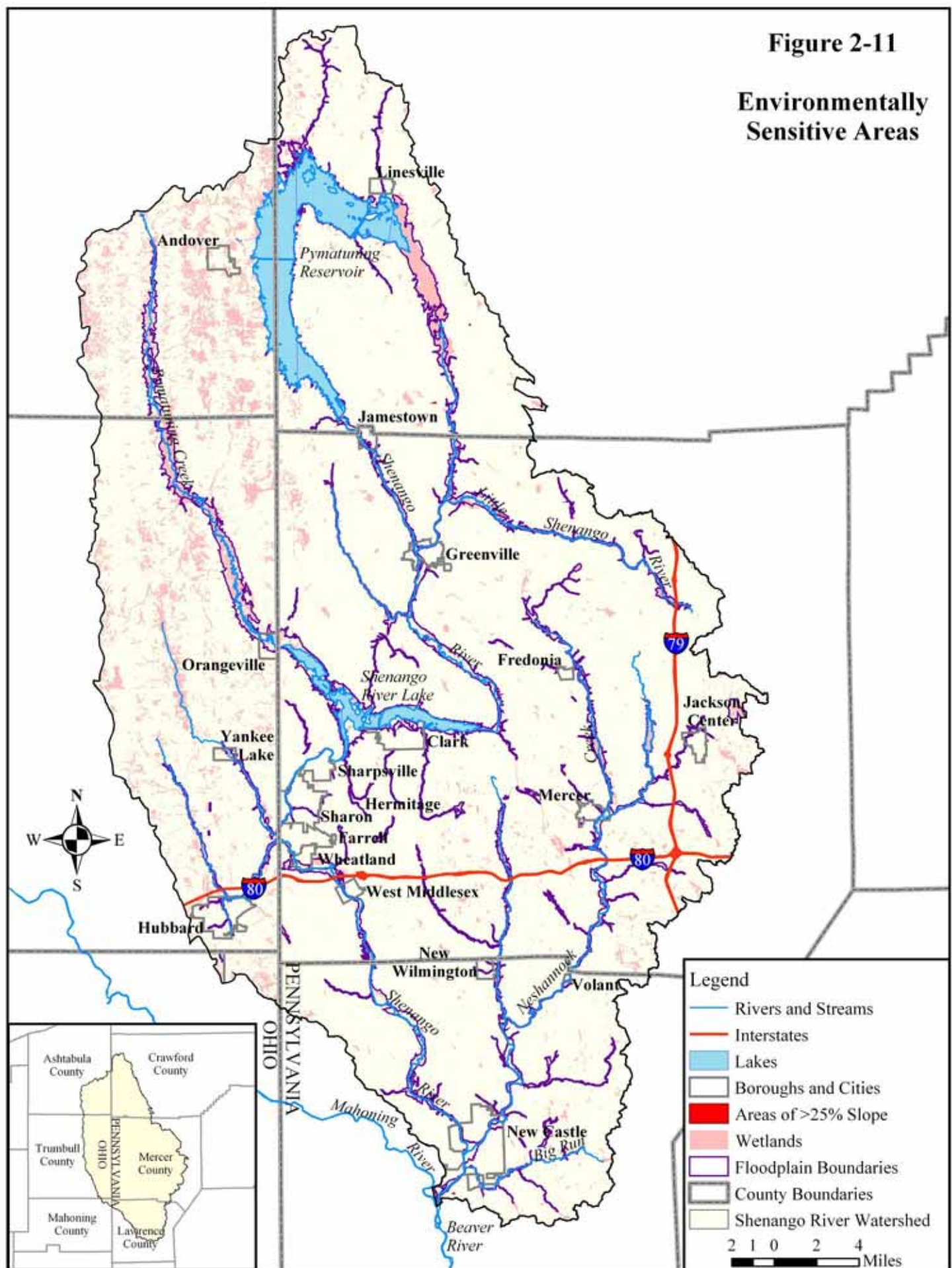
One of several wetlands in the Shenango River watershed

Erosion and Sedimentation

Erosion is the transfer of soil particles through air or water. The relocation of these particles is known as sedimentation. Erosion and sedimentation are natural earth-moving processes, but the extent of this movement can be greater than normal due to poor land-use practices. Erosion and sedimentation are discussed in greater detail in the Water Resources chapter.

Floodplains

A floodplain is the level land along the course of a river or stream that is formed by the deposition of sediment during periodic floods. Floodplains contain features such as levees, back swamps, delta plains, and oxbow lakes. These areas are critical to the waterway. Frequent flooding limits development. These areas often contain a unique diversity of plant and animal species. Floodplains in the watershed are discussed in the Water and Biological Resources chapters.



Wetlands

Wetlands are, “areas that are inundated or saturated by surface or ground water at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions” (USACE 2002). Wetlands are delineated according to hydrology, soil type, and vegetation. Whether man-made or naturally occurring, wetlands have a variety of appearances. Standing water, inundated soils, or an apparently dry field can be a wetland. More information about the wetlands located in the watershed can be found in the Water and Biological Resources chapters.

Fish and Wildlife Habitat

Habitats are the natural environments in which animals and plants reside. Healthy habitats are important to maintaining a diversity of biological resources. Interferences and changes to habitat affect the variety of plants and animals living there. Habitats where rare, threatened, and endangered species reside are critical to have in nature.

Important habitats within the watershed include riparian areas, forested areas, and wetlands. Riparian areas are lands located next to a body of water. When densely vegetated, they serve as a buffer against polluted runoff and provide habitat corridors for many varieties of wildlife. Forested areas, especially the non-fragmented forested areas, provide critical habitats for plants and animals. More information about fish and wildlife habitats in the watershed can be found in the Water and Biological Resources chapters.

Hazardous Areas

Illegal Dumpsites

In remote areas of the watershed, streambeds, hillsides, back roads, and coal mine refuse piles are inundated with old tires, appliances, and other items that people no longer want. These illegal dumps grow with continued use and can cause a variety of environmental and health impacts. Currently, it is the responsibility of each municipality to identify and clean up dumpsites.

PA CleanWays chapters throughout the state work to clean up and prevent illegal dumping through action and education. Local businesses, organizations, or clubs adopt rural roadways, trails, and/or waterways. These volunteers pick up trash in their adopted area two to three times a year, similar to the Adopt-a-Highway program run by Pennsylvania Department of Transportation.

Chapters of PA CleanWays are organized on a county basis. Lawrence County is the only county in the watershed with an active chapter. It facilitates the cleanup of areas affected by illegal dumping, addresses problem items, and educates the public about proper waste disposal and the environmental and personal consequences of improper disposal. The Lawrence County chapter has been involved in the recycling of electronics, household appliances, and tires. They have also hosted numerous cleanups and are currently surveying illegal dumpsites to identify, assess, and document every illegal dumpsite within the county.

Waste Sites

The **Comprehensive Environmental Response Compensation and Liability Act (CERCLA)**, commonly known as Superfund, was enacted in 1980 to provide broad federal authority to respond directly to releases of hazardous substances that may endanger public health or the environment. By creating a tax on the chemical and petroleum industries, a trust fund was established to provide for cleanup where no responsible party could be identified. In 1986, the Superfund Amendments and

Reauthorization Act (SARA) amended CERCLA. A listing of sites located within the watershed can be found in Appendix E.

Two kinds of action responses were identified in the law, short term and long term. Short-term removals require prompt response for releases or threatened releases. Long-term responses permanently and significantly reduce the dangers associated with releases or threats of releases of hazardous substances that are serious, but not immediately life threatening. These actions can be conducted only at sites listed on US EPA's National Priorities List (NPL).

Three sites have been identified on the NPL in the Shenango River watershed: River Road Landfill, Farrell Works Disposal Area of the Sharon Steel Corporation, and Westinghouse Electric Sharon Plant. All of these NPL sites are located in the Lower Shenango River subwatershed.

1. The River Road Landfill is a 102-acre inactive landfill that operated from 1962 until 1986. Municipal and industrial waste, including foundry and metal processing waste, polychlorinated biphenyl (PCB) wastes, asbestos, and residues from tank car cleaning were accepted at the site. Two sedimentation ponds catch runoff from the site and have controlled drainage into the Shenango River. The landfill was deleted from NPL on January 19, 2004.
2. Sharon Steel Corporation Farrell Works Disposal Area was added to the NPL in July 1998. The area is approximately 400 acres, located near the Pennsylvania/Ohio border and in the floodplain of the Shenango River. Around 1900, the area was used to dispose of blast furnace slag, electric arc furnace slag, basic oxygen furnace slag, and sludge. From 1949 to 1981, millions of gallons of spent pickle liquor acid were dumped over the slag, with hopes that the acid would partially evaporate and then be neutralized by the carbonates in the slag. Instead, groundwater was contaminated.
3. Westinghouse Electric Sharon Plant was added to the NPL in August 1990. This 58-acre facility manufactured electrical transformers from 1922 to 1985. Between 1936 and 1976, Westinghouse used blends of PCBs and trichlorobenzene in manufacturing some transformers. Contamination of soils, groundwater, and sediment in the nearby Shenango River were the result of spills on the land and leaks of solvent and oil from underground storage tanks. Phase III of the cleanup was expected to be completed in 2004.

The **Resource Conservation and Recovery Act (RCRA)**, a federal statute, regulates the transportation, handling, storage, and disposal of solid and hazardous materials. Regulatory responsibilities, including obtaining permits, identifying and listing hazardous waste, adhering to proper procedures when transporting or disposing of waste, developing risk management plans, and maintaining records, may be controlled by federal facilities. Requirements for underground storage tanks, including cover tank design, operation, cleanup, and closure, are also contained in RCRA.

There are 426 listed sites within the watershed with the majority of these sites being located in the Lower Shenango River subwatershed. A listing of all the sites is located in Appendix F.

The **Pennsylvania Hazardous Sites Cleanup Act (PA HSCA)** provides PA DEP with the capability to conduct cleanup activities at sites contaminated by hazardous substances. It also authorizes PA DEP to force the responsible parties to conduct cleanup actions or to repay public funds spent cleaning up the site. The majority of PA HSCA sites involve bankrupt facility owners, abandoned facilities, and inappropriate disposal of hazardous substances, not active facilities with financially viable owners. Funds from this act are also used to pay the state's share of federal Superfund cleanup actions in Pennsylvania.

There are five PA HSCA sites in the Shenango River watershed. They include the DuPont New Castle Junk Yard Site, Johnson Bronze, Carlson Mining Sanitation Landfill, CG Wood Site, and William Taylor Estate Disposal Site. The DuPont New Castle Junk Yard Site is the site of a former salvage yard. For a number of years, lead acid batteries were smashed on the ground and the lead plates were removed for sale to a smelter. Sampling at the site revealed that the soil and groundwater were contaminated with lead. Remediation efforts were completed in January 2000. Efforts to remediate the William Taylor Estate Disposal Site were initiated in 1989 with the landfill being properly closed in December 2000.

Brownfields

Brownfields are abandoned industrial sites that were contaminated during their operation. Although these sites are now vacant, the contamination remains. Brownfields are important areas because they can often be redeveloped after the contamination is mitigated. This is an important planning issue because the amount of remediation needed at a particular site must be determined before the redevelopment of a brownfield site is considered. Brownfield redevelopment is an important concept, because it allows abandoned land to be reused, which helps to reduce sprawl.

There are 16 brownfield sites identified in the Shenango River watershed. They are located in the Lower Shenango River subwatershed area, with one exception in the Middle Shenango River subwatershed. The brownfield sites are identified in Table 2-4. Remediation of a site is conducted to meet background, statewide health, or site-specific standards. A site is deemed in progress when US EPA receives a notice of intent to remediate and has not received a final report. Once a final report is received, the site is then transferred to the completed sites list.

Even though coal refuse piles and abandoned mines would fall under the popular definition of a brownfield, they do not qualify under Pennsylvania's present policy. Refuse piles and abandoned mines often lack the infrastructure needed for redevelopment. However, recent efforts to categorize abandoned mine sites as "greyfields" may lead to the development of a reuse program similar to brownfields.

Table 2-4. Brownfield Sites

Site Name	County	Municipality	Subwatershed	Cleanup Standard	Status
McDuff A	Lawrence	New Castle	Lower Shenango River	Site-specific	Completed
Caparo Steel *	Mercer	Farrell	Lower Shenango River	Statewide health	Completed
Caparo Steel AOC-3 Former Labor Ditch Area	Mercer	Farrell	Lower Shenango River	Statewide health	Completed
Caparo Steel AOC-9 Boilerhouse Fuel Storage	Mercer	Farrell	Lower Shenango River	Statewide health	Completed
Shenango Valley IDC Caparo Finishing Div B	Mercer	Farrell	Lower Shenango River	Statewide health	Completed
New Castle Refractories	Lawrence	New Castle	Lower Shenango River		In Progress
Armco Sawhill Tube *	Mercer	Sharon	Lower Shenango River		In Progress
Natl Castings Former	Mercer	Sharon	Lower Shenango River		In Progress
Nicholas Cianci Greenville Outparcels LLC	Mercer	Greenville	Middle Shenango River	Statewide health	Completed
Ben Pajank Estate	Mercer	Hermitage	Lower Shenango River	Statewide health	Completed
Roemer Industries	Mercer	Sharon	Lower Shenango River	Background	Completed
New Castle Industries	Lawrence	New Castle	Lower Shenango River		In Progress

* Indicates multiple facilities at the site.

Subsidence Areas

Subsidence is the downward movement of surface material involving little or no horizontal movement. Occurring naturally due to physical and chemical weathering of certain types of bedrock, subsidence usually occurs locally as a result of underground mining, excessive pumping of groundwater, or subsurface erosion due to the failure of existing utility lines. Subsidence usually occurs slowly over a long period of time, but can happen rapidly. The development of a sinkhole, for example, occurs when the support of the land is gradually removed over a period of time causing the land surface to sag and finally collapse, leaving a hole or cavity. Although subsidence is not common within the watershed, the potential for it still exists.

Earthquakes

Earthquakes in Pennsylvania and eastern Ohio are rare, but have occurred. In September 1998, a moderate earthquake was experienced in Jamestown, Pennsylvania near Pymatuning Lake. This earthquake was the biggest earthquake in Pennsylvania's recorded history with a magnitude of 5.2 on the Richter scale. As a result of the earthquake, approximately 120 household water supply wells went dry within three months after the earthquake. Eighty of these wells were on a ridge between Jamestown and Greenville, where water levels declined as much as 100 feet. As a result of the decrease of water in the ridge, the valley water levels increased approximately 62 feet.

Management Recommendations

Agricultural Lands

- Enroll agricultural landowners in the Conservation Reserve Enhancement Program (CREP), a voluntary program, to take marginal farmland out of production for wildlife habitat, or other similar programs.
- Identify additional local, state, and federal funding for the implementation of agricultural best management practices.
- Promote and implement conservation practices such as cover crops and crop residue, contour strips, grassed waterways, and minimal pesticide/herbicide use.
- Promote and utilize farmland preservation programs to sustain agricultural base and rural heritage.
- Promote the importance and economic viability of small farms through marketing education and creation of avenues for adding value to locally produced agriculture commodities.
- Protect active farmlands to retain their agricultural uses by designating them as agricultural security areas, purchasing conservation easements, or assisting in the multi-generation transfer of ownership.
- Promote local production of agricultural products, producers, markets, and preservation programs.

Erosion and Sedimentation

- Establish land-use planning and zoning to limit development in floodplains and other critical areas subject to erosion and sedimentation problems.
- Include sound geologic investigation and best management practices during maintenance and construction of roadways.
- Utilize best management practices to control erosion and sedimentation in agriculture, forestry, development, and mining industries.

Forestry

- Host workshops and/or programs promoting sustainable forestland management for loggers, landowners, and municipal officials.
- Decrease forest fragmentation by maintaining contiguous forest tracts and/or by maintaining travel corridors between non-contiguous forest tracts.
- Decrease forest parcelization by protecting large forest tracts with conservation easements, forest stewardship program participation, and zoning ordinances.
- Educate forestland owners by providing them with accurate information regarding sound silviculture practices, forest management plan development, and insect and disease problems that can affect forest health.
- Encourage the development and use of Forest Stewardship Plans or forest management plans and participation in the PA Forest Stewardship Program and/or the Tree Farm Program.
- Discourage the use of high grading practices like diameter limit harvests and selective cutting, and encourage timber harvesters to use sustainable best management practices under the direction of a professional forester.
- Promote tree plantings and sustainable harvesting.
- Work with Woodland Owners Association to educate the public, restore degraded areas, and develop demonstration areas.
- Support laws and regulations to maintain whitetail deer populations at levels that will ensure healthy forests, productive agricultural lands, and healthy deer populations.

Waste Disposal

- Educate citizens about the economic and environmental impacts of illegal dumping.
- Partner with local landowners, businesses, and community groups to sponsor community cleanups to remove trash along roadways, streambanks, and at dumpsites.
- Work with county planning departments to prohibit any new or proposed landfills near water sources and residential areas.
- Establish chapters of PA CleanWays in Mercer and Crawford counties to assist in identification, education, and cleanup of illegal dumpsites.
- Identify additional local, state, and federal funding to address the waste disposal needs of the area.
- Conduct an inventory and develop a remediation plan for illegal dumpsites in the watershed.
- Educate citizens about traditional and innovative ways to reduce, reuse, recycle, and properly dispose of household hazardous waste by providing public service announcements, recycling workshops, and other education and outreach programs.
- Provide convenient and affordable alternative disposal options, such as offering special collection days and drop-off locations for appliances and household hazardous wastes.

Riparian Corridors

- Educate landowners along waterways about the value of riparian buffers.
- Establish greenway corridors and trails along waterways.
- Establish and protect riparian buffers along streams using smart growth practices as identified in the Project Area Characteristics chapter.
- Continue streambank restoration and riparian buffer initiatives on agricultural lands to minimize nutrients and sediments entering the waterways.

Development and Redevelopment

- Establish a planned approach to growth; revitalize existing structures before developing new lands.
- Promote the inclusion or preservation of open space in community development programs.
- Establish more greenways and forestlands that are available for public use.
- Protect natural areas by converting them into parks, purchasing conservation easements, or utilizing other conservation methods.
- Promote redevelopment of abandoned industrial sites through the brownfields program, incentives, tax breaks, or other efforts.
- Redevelop the reclaimed River Road Landfill site for a park/recreation area.

Other

- Conduct outreach campaigns to educate watershed residents about the land uses in their communities.
- Establish a regional sanitary authority.
- Educate the public to utilize practices such as "Leave no trace" on public lands and private lands open to public use.

CHAPTER 3. WATER RESOURCES

A watershed is a basin-like landform defined by high points and ridgelines that descend into lower elevations and stream valleys. A watershed carries water "shed" from the land after rain falls and snow melts. Drop by drop, water is channeled into soils, groundwater, creeks, and streams, making its way to larger rivers, and eventually the sea. Water is a universal solvent capable of dissolving and transporting many chemicals. What we put on the ground—lawn chemicals, agricultural fertilizers, salt on roads in winter, oils from exhaust on highways—affects water quality downstream. When we remove vegetation from the watershed, we remove nature's mechanisms for storing and cleaning water. Asphalt surfaces, rooftops, roads, and parking lots keep water from reaching soils. Rain is piped away before soils can retain it, increasing the likelihood of flooding and erosion. Land development and stormwater management practices can be adapted to have fewer negative impacts on environmentally sensitive areas if the characteristics of natural systems are considered.

The pollutants that come from runoff contribute to non-point source pollution, the leading cause of water quality impairments in Pennsylvania. Non-point source pollution comes from many diffuse sources and is caused by rainfall or snowmelt moving over and through the ground. As the runoff moves, it picks up and carries away natural and human-made pollutants, finally depositing them into streams, lakes, rivers, wetlands, and underground sources of drinking water.

All the factors that go into making a stream, as well as the diversity of in-stream habitats, are extremely complex and interconnected. Factors affecting a stream's composition include: precipitation (seasonal variations), topography (determines stream gradient), geologic substrate (from which minerals are leached), land uses (determine sediment and contaminant loading of surface runoff and groundwater), soil and bedrock types (determine groundwater availability), sunlight or shade (affect temperature and algae growth), and riparian vegetation (for shade, nutrient source, insect habitat, and more). In addition, streams widen and their volumes increase as tributaries and other streams join them. The Shenango River watershed has been shaped by glaciation, resulting in a dendritic drainage pattern and a significant number of lakes (reservoirs), ponds, and wetlands that now characterize the landscape.

How the Water Spirit Got Her Power

Once, long ago, Sun was the ruler of all the earth. Next to him, the other spirits were as the sparrow beside the grizzly bear. So the spirits had a secret meeting and elected the water spirit to approach the Sun to give up some of his power.

Water went to Sun, and formed a clear, deep pool at his feet. When Sun saw his own face reflected in the pool, he was so delighted that he promised water anything she wanted. When she demanded some of his power, he realized that he had been tricked, but according to his word, he gave power to all of the other spirits. Water, for her part, got more than anyone, and became, next to Sun, the most powerful force on earth.

*-From a plaque next to Takaklaw Falls,
Yoho National Park, British Columbia,
Canada*



A portion of the Shenango River watershed

Only 30 to 40 percent of the rain or snow that reaches the ground goes directly to streams. Most of it, surprisingly, is taken up and used internally by plants. Some water infiltrates soils and moves below the surface as groundwater, feeding plants and replenishing aquifers. Water exposed to the air evaporates from streams, rivers, wetlands, and plants. It returns to the atmosphere to fall again as precipitation. This “water cycle” cools the planet, cleans the air, and sustains life as we know it.

Pennsylvania is second only to Alaska in its abundance of running waters, containing approximately 83,161 miles of rivers and streams. The Pennsylvania Department of Environmental Protection (PA DEP) has developed minimum water quality standards (Pennsylvania Code 1997) for these streams based on their designated uses. Streams are protected for their aquatic life, water supply, recreation and fish consumption, high quality or exceptional value, or navigation uses. Shenango River watershed streams and their designated uses are discussed further in the Water Quality section, and Table 3-3.

According to the Ohio Environmental Protection Agency (OhioEPA), beneficial use designations describe existing or potential uses of waterbodies. They take into consideration the use and value of water for public water supplies, protection and propagation of aquatic life, recreation in and on the water, agricultural, industrial, and other purposes. Examples of beneficial use designations include: public water supply, primary contact recreation, and aquatic life uses (warm-water habitat, exceptional warm-water habitat, etc.). For more information about the definitions of Ohio waterway use designations, refer to the Ohio Administrative Code website found in Appendix N.

The intended purpose of this conservation plan is to supplement and expand upon the information contained in the Interstate Pymatuning/Shenango Watershed Plan. A great deal of information about the water resources within the Shenango River watershed is contained in this chapter; however, some additional information can be found in the Interstate Pymatuning/Shenango Watershed Plan.

Major Tributaries

Tributaries are streams that flow to larger streams or other bodies of water. Major tributaries of the Shenango River are listed in Table 3-1, and shown in Figures 3-1, 3-2, 3-3, and 3-4. These streams form a dendritic drainage pattern on the landscape of the Shenango River watershed.

In addition to the impact that streams and rivers have on the natural landscape, humans have also made an impact on the landscape and course of water flow. One such example is the Erie Extension Canal. The Erie Canal crossed New York State from Albany to Lake Erie, and the Pennsylvania Main Line Canal crossed southern Pennsylvania from Philadelphia to Pittsburgh. The Erie Extension Canal was built to connect these two canals and, essentially, the Ohio River and Lake Erie. Irish laborers dug the entire canal, which is 130 miles long. Although the Erie Extension Canal was no longer used after the destruction of an aqueduct across Elk Creek Gorge in 1873, remnants of it still remain.



*Trace of the old canal in Greenville
photo courtesy of Nathan Leary, 1999*

Table 3-1. Major Tributaries

Tributary	Watershed Area (sq. mi.)	Approx. % of Drainage
<i>Upper Shenango Subwatershed</i>	155	14%
Linesville Creek	9.73	
Bennett Run	6.99	
Paden Creek	16.80	
Gravel Run (OH)	11.60	
Black Creek (OH)	unknown	
McMichael Creek (OH)	unknown	
<i>Middle Shenango Subwatershed</i>	434	41%
Sugar Run	8.67	
Little Shenango River	109.00	
Morrison Run	5.21	
Crooked Creek	45.60	
Bluff Run	3.90	
McDowell Run	3.75	
Randolph Run	6.48	
Unger Run	2.21	
Mathay Run	5.77	
Big Run	26.50	
Lawango Run	5.09	
Lackawannock Creek	9.01	
Daley Run	3.12	
Magargee Run	9.08	
Golden Run	4.51	
Brush Run	5.49	
Pymatuning Creek	171.00	
Stratton Creek	unknown	
Sugar Creek	unknown	
Mill Creek	unknown	
Booth Run	10.50	
Pine Hollow Run	4.38	
<i>Lower Shenango Subwatershed</i>	208	19%
McCullough Run	8.18	
Thornton Run	1.22	
Big Run	4.17	
Pine Run	2.48	
Little Yankee Run	41.79	
Deer Creek	1.28	

Table 3-1. Major Tributaries (continued)

Tributary	Watershed Area (sq. mi.)	Approx. % of Drainage
Bobby Run	2.92	
Hogback Run	8.03	
Turkey Run	2.93	
Buchanan Run	6.99	
Deer Creek	13.80	
<i>Neshannock Creek/Big Run Subwatershed</i>	270	26%
Neshannock Creek	242.00	
Otter Creek	45.90	
Mill Run	4.22	
Oilmill Run	unknown	
Cool Spring Creek	50.80	
Mill Creek	11.30	
Yellow Creek	21.70	
Fox Run	8.23	
Beaver Run	3.41	
Pine Run	8.47	
Mill Run	5.20	
Indian Run	6.72	
Potter Run	5.63	
Little Neshannock	50.60	
West Branch	17.40	
Hottenbaugh Run	13.90	
Big Run	28.50	

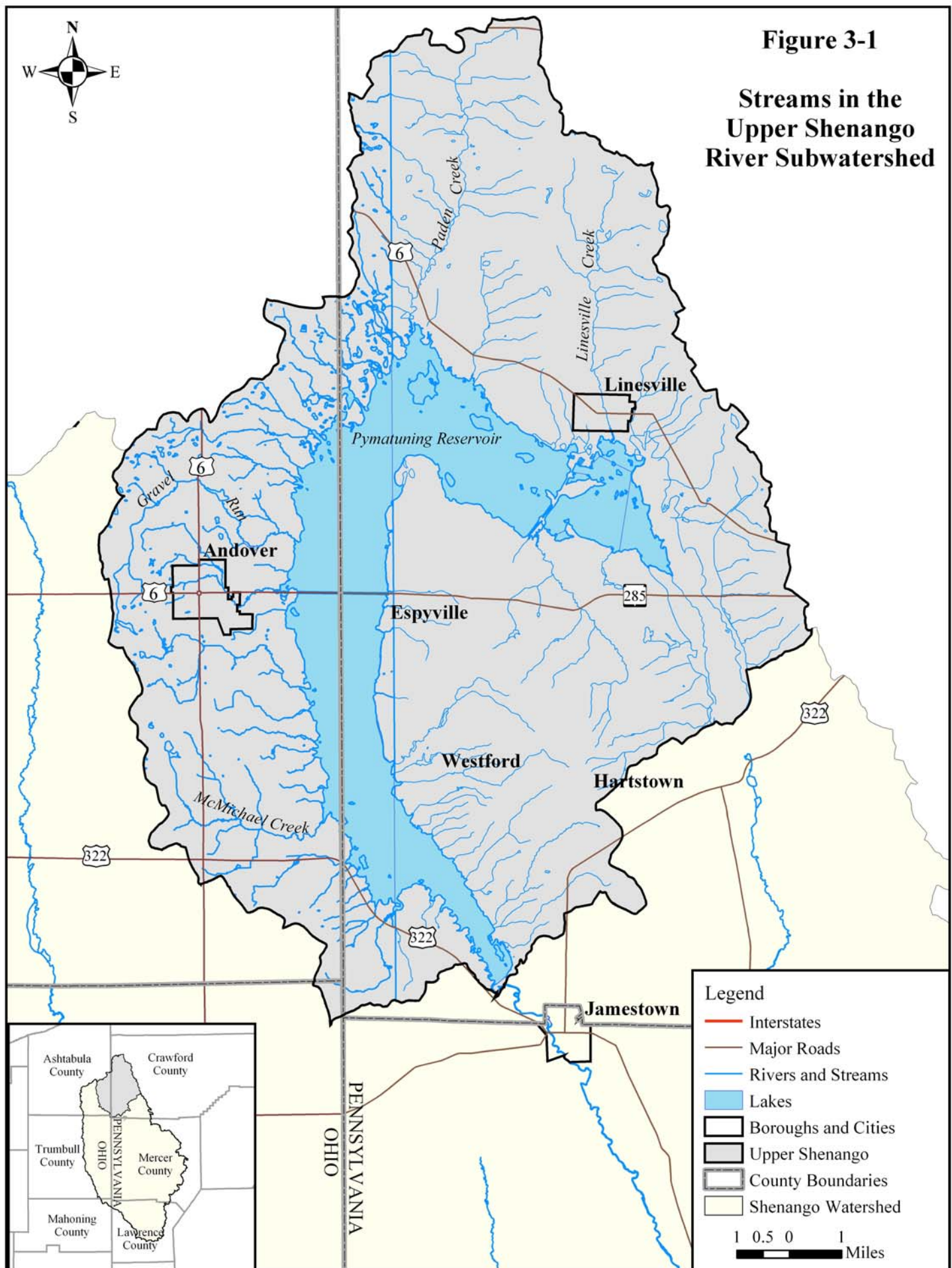
Wetlands

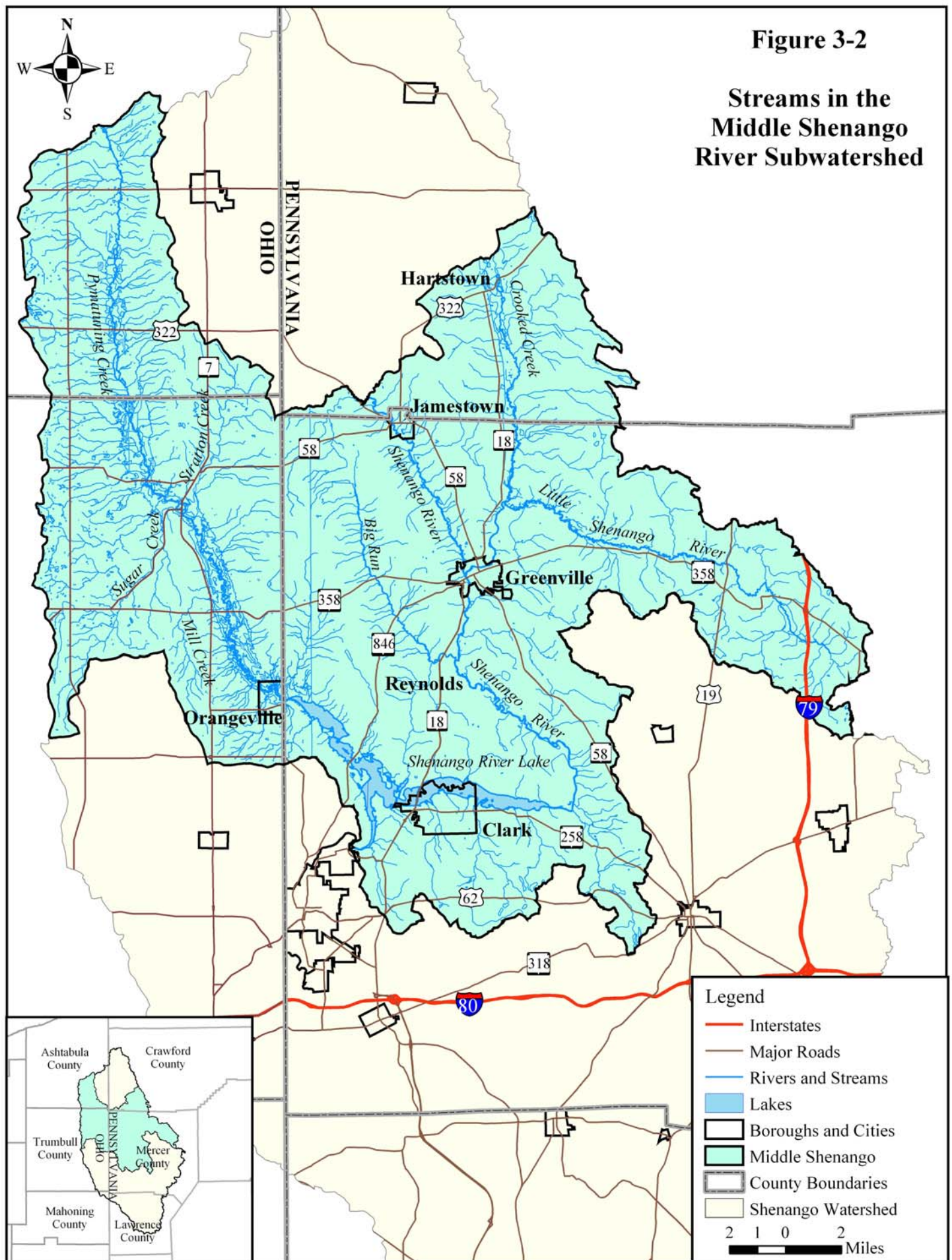


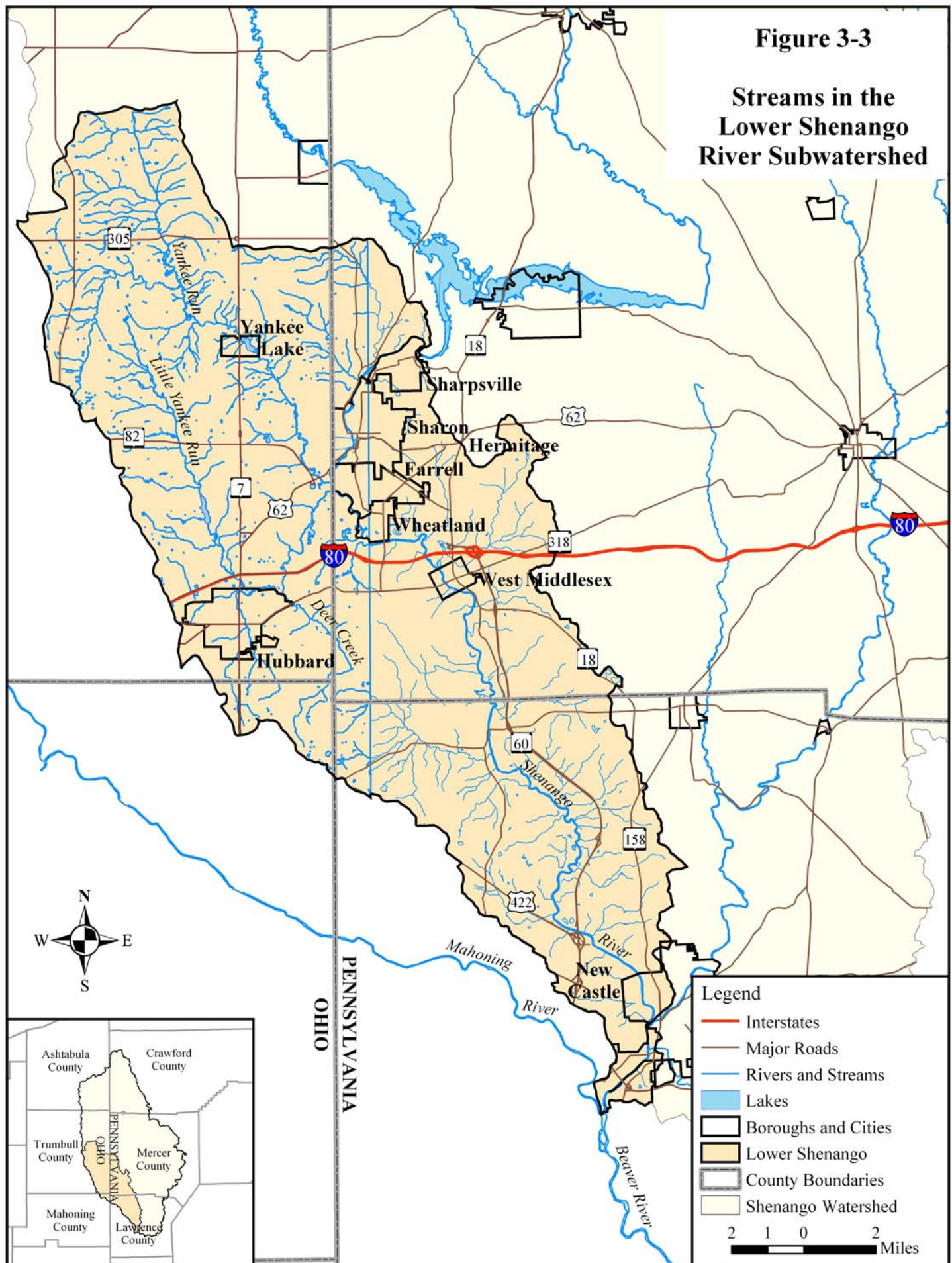
Wetland area at Black Jack Swamp

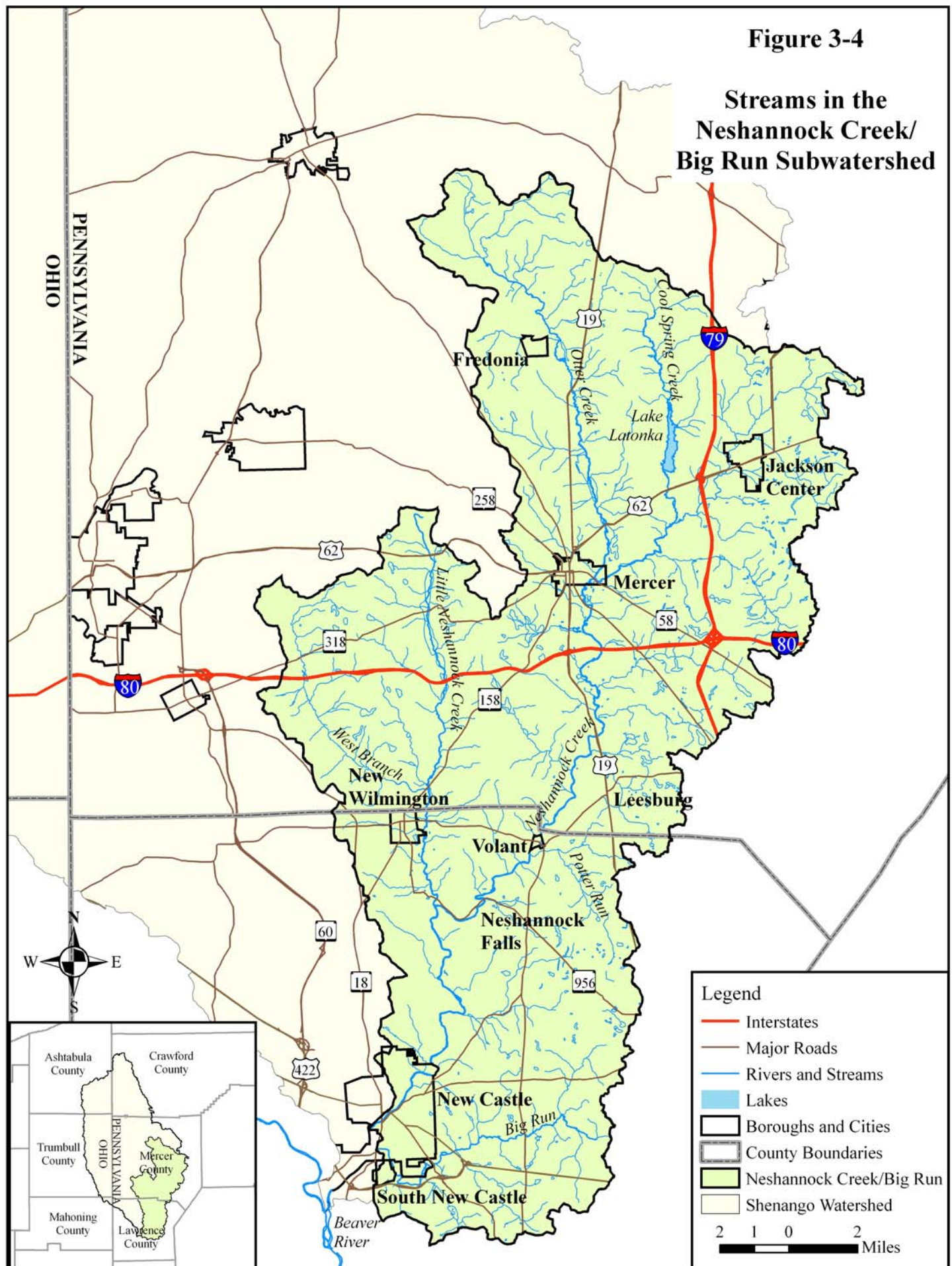
Wetlands can be defined as marshes, bogs, swamps, wet meadows, or shallow ponds (further described in the Biological Resources chapter). To be considered a wetland, an area must have supporting hydrology, vegetation, and soils. Frequently occurring in low areas, wetlands are regulated by the Pennsylvania Dam Safety and Encroachments Act – 25 PA Code, Chapter 105 – which reviews all water-related activities to protect and conserve the natural resources of the commonwealth.

United States Environmental Protection Agency (US EPA) and United States Army Corps of Engineers (USACE) use the 1987 Corps of Engineers Wetlands Delineation Manual to define wetlands for the Clean Water Act Section 404 permit program. Section 404 requires a permit from the USACE or









authorized state for the discharge of dredged or fill material into the waters of the United States, including wetlands. In addition, the Food Security Act of 1985 included a Swampbuster provision that discouraged the removal of wetlands for agricultural purposes by denying federal farm program benefits for producers who converted wetlands after December 23, 1985 (Tiner 1999).

Wetlands make important contributions to the health of aquatic systems on a watershed basis by purifying water, filtering runoff, abating floods, and decreasing erosion. In addition, wetlands provide habitat for countless numbers of plants and animals including over 40 percent of all federally listed threatened or endangered species. The functions of wetlands are very important to water quality and quantity, as well as land conservation. Wetlands help reduce soil runoff and limit the effects of non-point source pollution from surrounding areas. Plants found in wetlands help slow the movement of water, allowing the sediment to drop out. Nutrients, such as nitrogen and phosphorus, which cling to the soil particles, are deposited in the wetland where they are used for food and energy. Chemicals, such as pesticides and herbicides, may be trapped in the wetland and broken down by sunlight and microorganisms. Wetlands also prevent downstream flooding by acting as sponges to absorb heavy rainfall.

Approximately 1.4 percent, or 403,924 acres, of Pennsylvania's land surface is covered by wetlands. Wetlands are most abundant in the glaciated portions of northeastern and northwestern Pennsylvania. Wetlands cover at least five percent of the entire Shenango River watershed (28,949 acres in Ohio and 39,226 acres in Pennsylvania). The wetlands in each of the subwatersheds are shown in Figures 3-5, 3-6, 3-7, and 3-8.

Wetland Loss

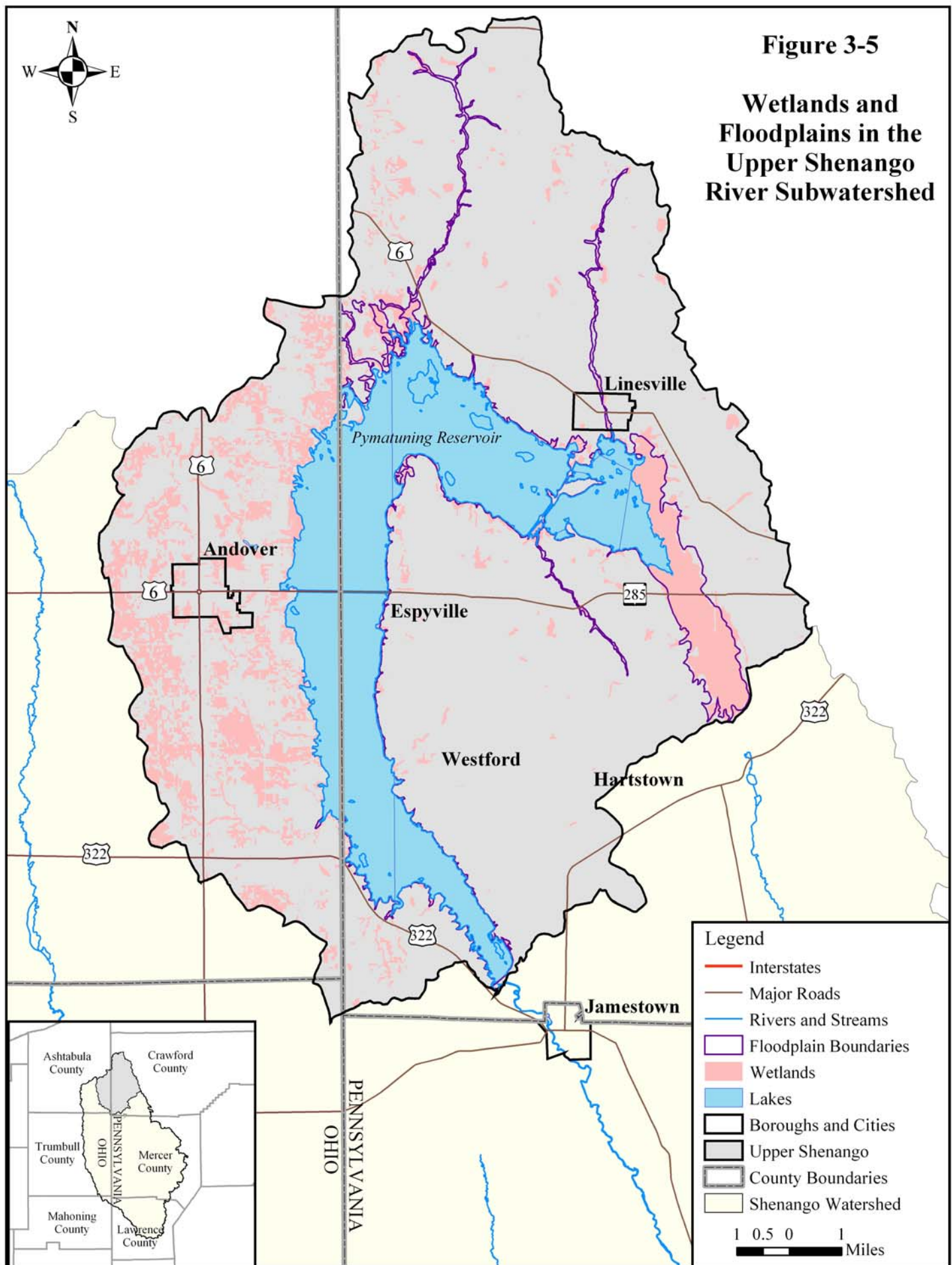
Wetland loss, or the loss of wetland area via the conversion of wetlands to non-wetland land use, from human and natural disturbances is a key issue. In the past 200 years, over 56 percent of wetlands in Pennsylvania have been filled or destroyed (Dahl 1990), and nearly 90 percent of wetlands in Ohio have disappeared (<http://www.dnr.state.oh.us/wetlands/history.htm>). Specific statistics for historic wetland loss within the Shenango River watershed are currently not available.

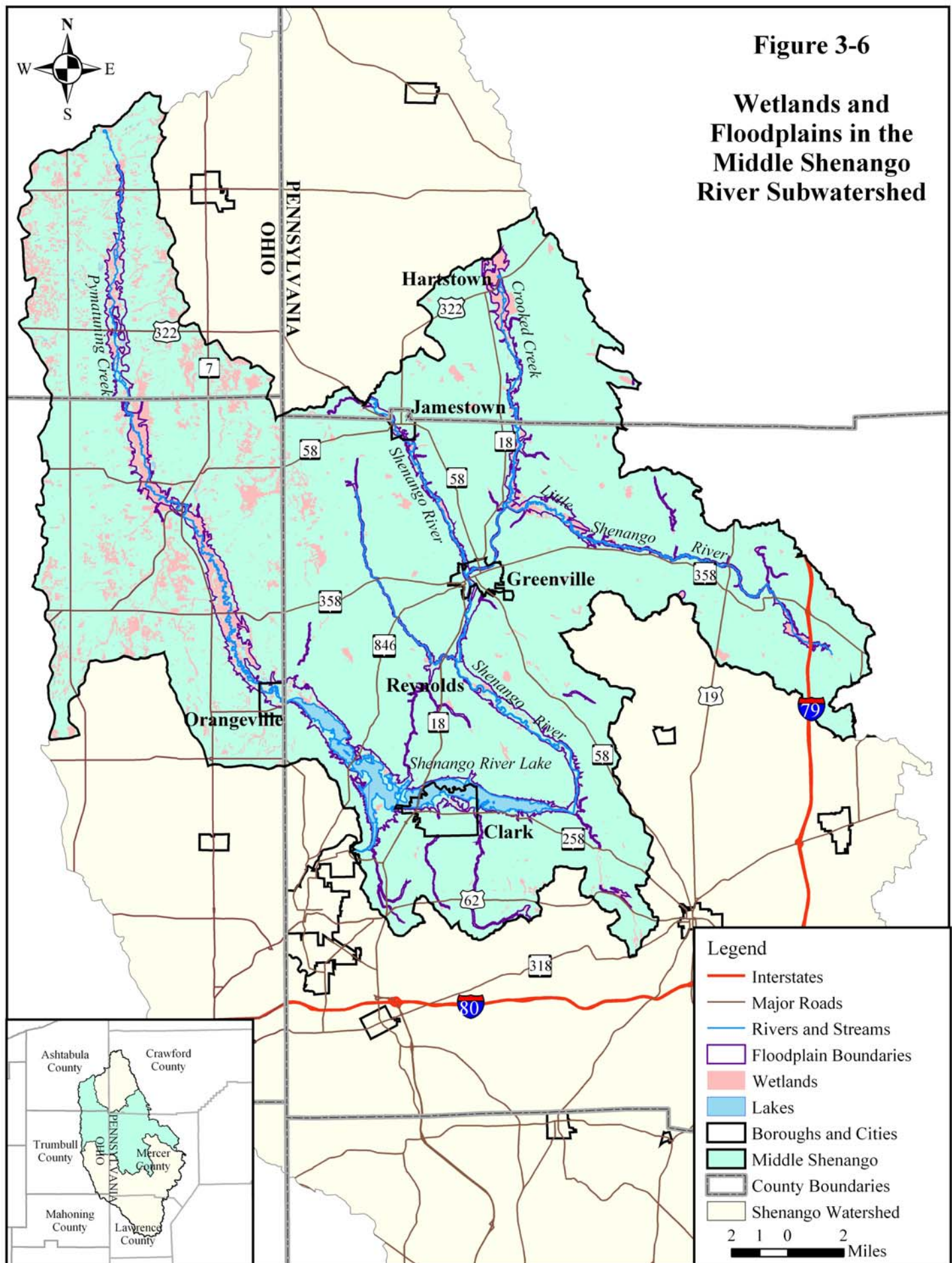
The National Wetland Loss Index indicates that the Shenango River watershed experienced a high level of wetland loss between 1982 and 1992 (US EPA). A high level of loss means that according to the Natural Resources Conservation Service (NRCS), on rural, non-federal lands, wetland loss has been greater than two percent between 1982 and 1992. This loss level is also determined by the U.S. Fish and Wildlife Service on federal and non-federal lands, which indicates historic percent loss from the 1780s to the 1980s was greater than 70 percent.

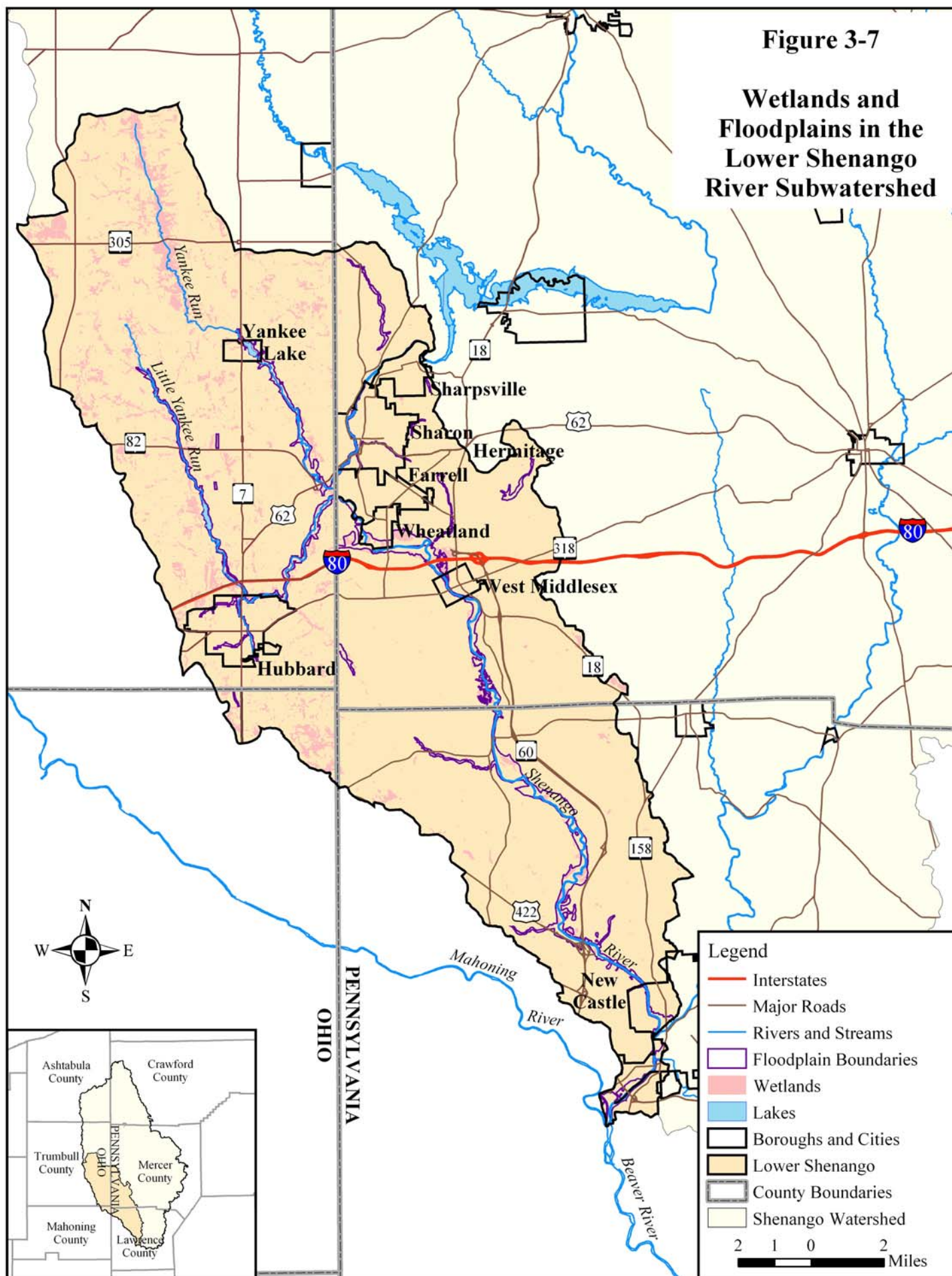
According to a U.S. Department of Agriculture (USDA) NRCS study, the major causes of wetland loss in the northeastern United States, including Pennsylvania, during the 1990s were primarily from development (67.2%), forestry (16.3%), agriculture (9.0%), and other causes (7.5%). The Midwest region of the United States, including Ohio, reports that during the 1990s wetland loss was caused by agriculture (51.9%), development (28.7%), forestry (19.3%), and other causes (0.1%) (NRCS 2000). Historically, conversion of wetlands to agricultural land uses has been the dominant reason for wetland loss. After 1982, development became the primary contributing factor to wetland loss (Petrie et al. 2001).

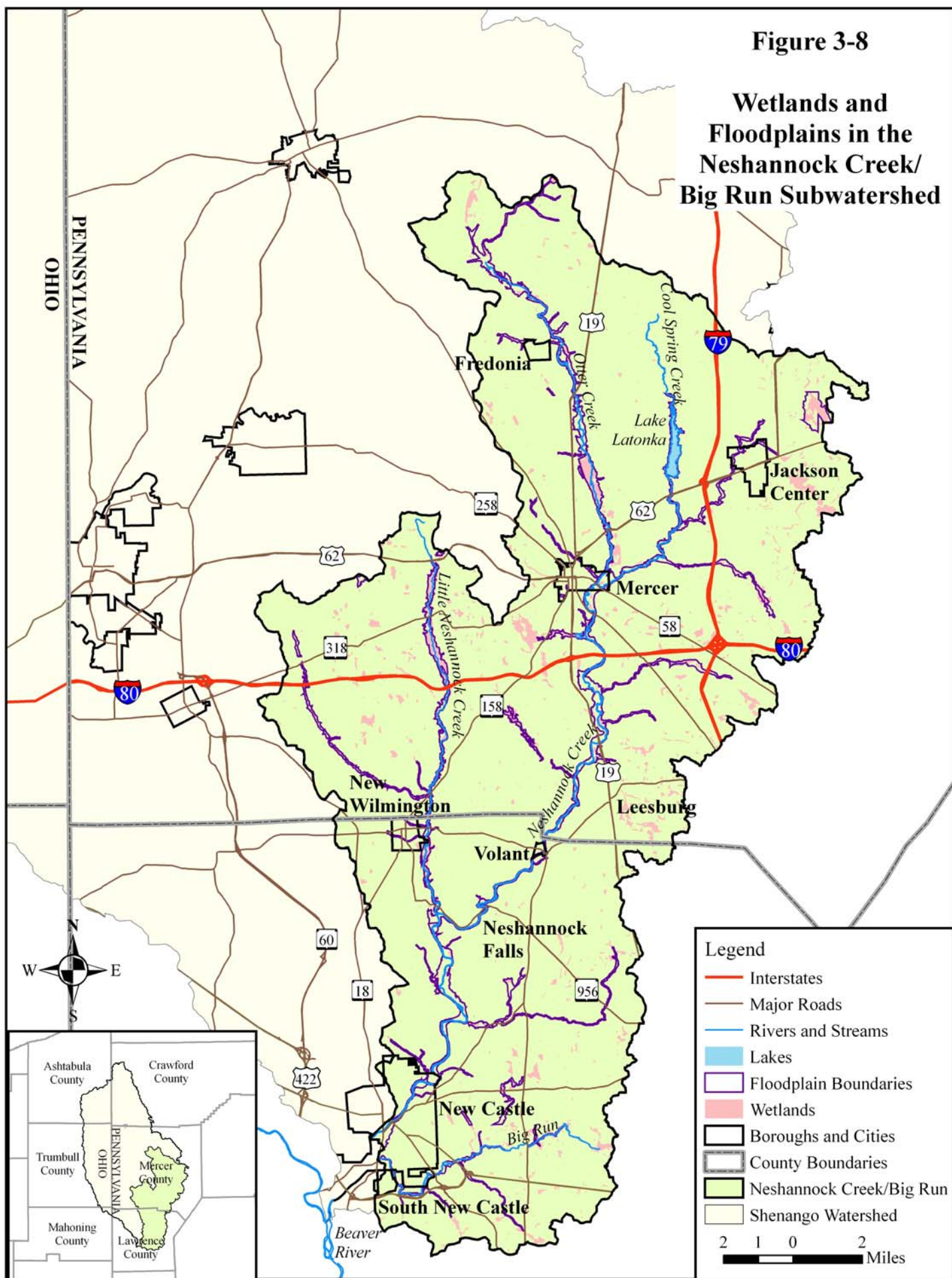


*Headwaters of Paden Creek,
tributary to Shenango River*









Floodplains

Floodplains are another feature of a stream that provide an array of benefits to watershed systems, including:

- Gradual retention and release of groundwater, overland flows, and flood flows;
- Surface and groundwater filtration;
- Sediment deposition; and
- Production of food sources, cover, and thermal protection for organisms living in riparian or floodplain areas.



A Shenango River tributary utilizing its floodplain

Floodplain alterations, such as the removal of vegetation and encroachment by residential and commercial development, interrupt the natural relationship between the stream and its adjacent floodplain. Habitat and food for organisms dwelling within the riparian or floodplain area is also compromised when encroachment or vegetative removal occurs. Adequate vegetation and floodplain integrity, in conjunction with restoration and conservation practices, are key factors in limiting degradation of the water quality and biological resources and protecting those living downstream.

The National Flood Insurance Act of 1968 and Flood Disaster Protection Act of 1973 were implemented to handle issues of floodplain alterations and subsequent watershed flooding. The programs were expanded through the National Flood Insurance Reform Act of 1994 and serve as a foundation for the National Floodplain Insurance Program (NFIP), which assists in community floodplain and flood insurance planning through the implementation of local floodplain management ordinances.

In addition, the federal government published a handbook of federal programs offering non-structural flood recovery and floodplain management alternatives in 1998. In response to severe flooding problems, such as loss of life and property, natural resources, and functional floodplains, the federal government has shifted its focus to improve its floodplain management. This shift in focus entails utilizing non-structural approaches to flooding, as opposed to the traditional structural approaches such as dams, levees, and channels that serve to control flooding. Non-structural approaches tend to modify the susceptibility to flooding. Examples of non-structural approaches include property acquisition, relocation, elevation, flood-proofing of existing structures, rural land easements, and restoration of wetlands. By utilizing the non-structural approach to floodplain management, communities will be made safer and stronger by reducing flood losses, diminishing the mental, physical, and economic toll on towns and farms that have suffered repeated damage, cutting long-term costs, and helping to restore lost or impaired environmental resources. The handbook can be found at <http://www.fema.gov/pdf/hazards/ombflood.pdf>.

All of the Pennsylvania municipalities within the watershed have floodplain ordinances; however, many of them are over 20 years old and should be updated according to land-use changes that have occurred since that time. Table 3-2 shows the dates that municipalities entered the national flood program, which requires that an ordinance be adopted. Should a homeowner have concerns about whether they live in a floodplain, or if they are subject to a floodplain ordinance, they should contact their local municipality. Figures 3-5, 3-6, 3-7, and 3-8 indicate floodplain areas by subwatershed.

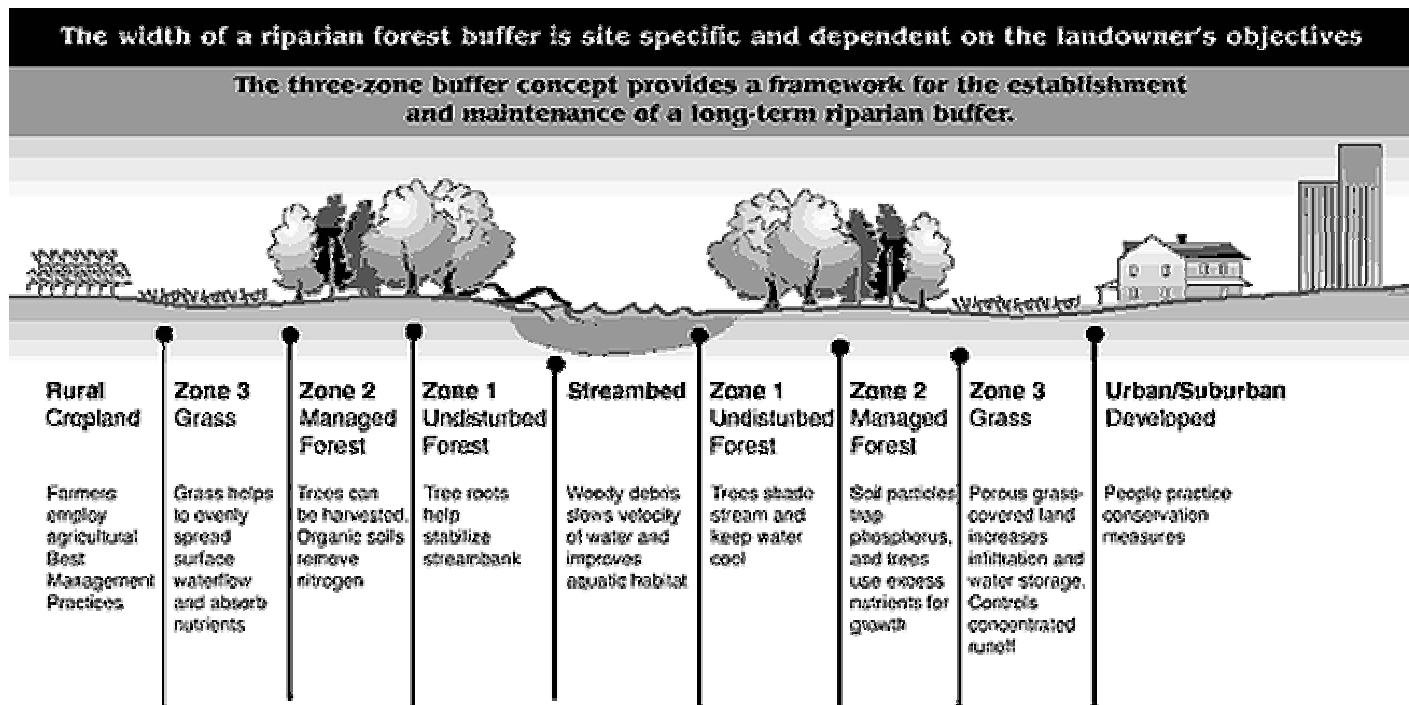
Table 3-2. Municipalities Participating in the National Flood Insurance Program

Municipality	Date of Entry	Municipality	Date of Entry
<i>Crawford County</i>		<i>Mercer County (continued)</i>	
Beaver Township	9/10/1982	Findley Township	2/4/1983
Conneaut Township	9/10/1984	Fredonia Borough	11/17/1978
East Fallowfield Township	9/24/1984	Greene Township	6/30/1976
Linesville Borough	6/5/1985	Greenville Borough	7/16/1981
North Shenango Township	10/1/1986	Hempfield Township	2/15/1991
Pine Township	6/19/1985	Hermitage	9/30/1981
Sadsbury Township	8/19/1991	Jackson Center Borough	6/18/1982
South Shenango Township	7/3/1985	Jamestown Borough	9/10/1982
Summit Township	5/19/1987	Jefferson Township	6/1/1986
West Fallowfield Township	7/3/1985	Lackawannock Township	6/30/1976
West Shenango Township	2/1/1985	Lake Township	6/18/1982
<i>Lawrence County</i>		Mercer Borough	3/15/1977
City of New Castle	8/15/1978	New Vernon Township	10/15/1982
Hickory Township	8/2/1983	Otter Creek Township	12/1/1986
Mahoning Township	2/18/1983	Perry Township	12/17/1982
Neshannock Township	5/17/1982	Pine Township	2/25/1983
New Wilmington Borough	8/3/1984	Pymatuning Township	6/1/1989
Plain Grove Township	8/3/1984	Salem Township	5/1/1986
Pulaski Township	12/31/1982	Sandy Creek Township	10/1/1986
Scott Township	11/1/1986	Sandy Lake Township	9/3/1982
Shenango Township	4/3/1978	Sharon City	10/17/1978
Slippery Rock Township	11/1/1986	Sharpsville Borough	5/25/1978
South New Castle Borough	7/31/1978	Shenango Township	9/4/1991
Taylor Township	8/3/1984	South Pymatuning Township	3/18/1991
Union Township	11/5/1982	Springfield Township	7/16/1982
Volant Borough	9/28/1979	Stoneboro Township	3/18/1991
<i>Mercer County</i>		Sugar Grove Township	9/17/1982
Clark Borough	7/30/1982	West Middlesex Township	9/4/1991
Coolspring Township	9/17/1982	West Salem Township	1/21/1983
Delaware Township	7/30/1982	Wheatland Borough	2/15/1978
East Lackawannock Township	7/23/1982	Wilmington Township	2/4/1983
Fairview Township	1/1/1987	Wolf Creek Township	6/25/1982
Farrell City	4/17/1978	Worth Township	2/4/1983

Riparian Buffers

Riparian buffers are strips of grass, shrubs, and/or trees along the banks of rivers and streams that filter polluted runoff and provide a transition zone between water and human land use. Buffers are complex ecosystems that provide habitat and improve the stream communities they shelter. Figure 3-9 depicts the three-zone buffer concept, which shows how various land uses within riparian corridors can affect the benefits of the buffer. Riparian buffers work to protect our waterways in many ways:

Figure 3-9. Three-Zone Buffer Concept



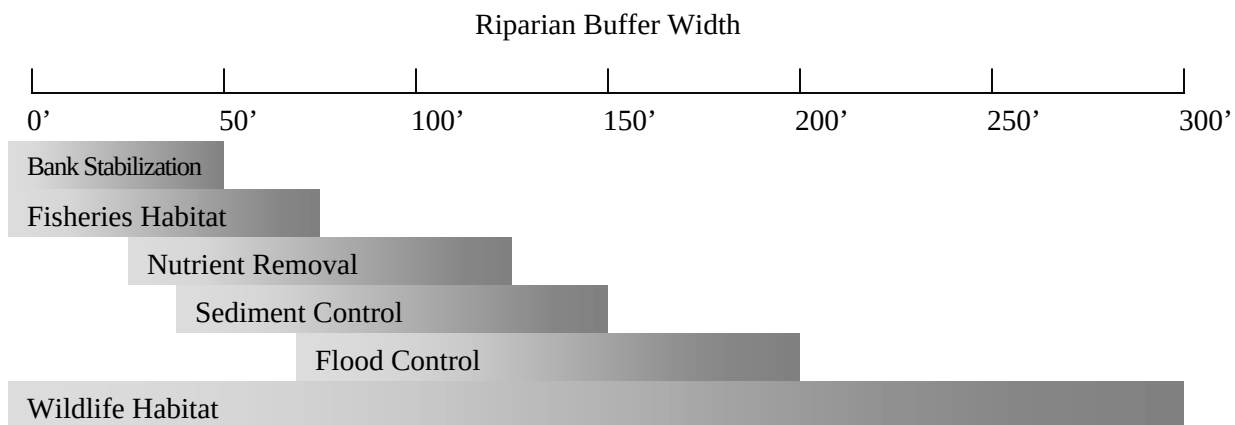
Source: Maryland Cooperative Extension

- **Filter sediment.** Buffers help to trap sediment and debris from surface runoff. Vegetation within the buffer can absorb up to 100 percent of the nutrients in the sediment, depending on its width and characteristics.
- **Filter, transform, and collect pollutants.** Fertilizer and animal wastes contain pollutants, such as excess phosphorus and nitrogen, which bond to soil particles and can be trapped by riparian buffers, preventing them from washing into surface and groundwater. Chemical and biological activity in the soil can capture and transform nitrogen and other pollutants into less harmful forms. Buffers also act as a sink, taking up excess water and nutrients by their root systems and storing them in the biomass of the plant.
- **Regulate stream flow.** Riparian buffers allow water to infiltrate the soil and recharge the groundwater supply by slowing the velocity of the runoff. Since groundwater will reach a stream or river at a much slower rate than surface runoff, buffers help to control flooding during heavy precipitation events and maintain stream flow during the driest time of the year.
- **Stabilize streambanks.** The roots of the buffer vegetation help to stabilize streambanks by holding the soil together. Stems and leaves of the vegetation protect banks by deflecting precipitation and the force of the stream's current.
- **Stabilize streambeds.** Riparian buffers can also reduce the amount of streambed scour by absorbing surface water runoff and slowing water velocity. When vegetation is absent from the riparian area, more surface water reaches the stream, causing the water levels to rise more quickly during heavy rainfalls and snowmelt. This stronger flow can scour streambeds, disturbing the aquatic life there.
- **Provide wildlife habitat.** The unique habitat found in riparian buffers is home to a great diversity of plant and animal species. Some of these species, in fact, are only found within riparian areas. Continuous stretches of riparian buffer also serve as travel corridors for wildlife.

- *Provide food and aquatic habitat.* Forested riparian buffers offer an array of benefits to aquatic habitat. They improve the quality of the water by shading, filtering, and moderating stream flow. Shading is important to maintain cooler temperatures, which hold more oxygen, and reduce stress on fish and aquatic insects. Woody debris and leaves from forested riparian areas feed the aquatic food web, and can also create pools and riffles that provide cover and habitat for fish and aquatic insects. Fallen trees and limbs can also reduce erosion by slowing flow.
- *Enhance recreation and aesthetics.* Riparian buffers can create important greenways along streams and rivers by blocking views of nearby development and allowing privacy for riverfront landowners. Recreational activities, such as hiking trails and camping, are also a possibility with riparian buffers.

The intended purpose of the riparian buffer should determine its width. Topography, hydrology, soil type, and vegetation should be considered when determining the width. Figure 3-10 shows the buffer widths recommended to achieve various protections.

Figure 3-10. Recommended Riparian Buffer Widths



An effort currently underway within the Shenango River watershed is the Riparian Restoration and Protection Initiative. This project has identified the watershed as a priority, based on the following factors:

- Impairment, as identified by PA DEP;
- Watersheds that feed into lakes that are important drinking water supplies, have recreational amenities, or are storehouses of biological diversity;
- Impaired by agriculture and having highly erodible soils;
- High Quality or Exceptional Value designation; and
- Watersheds that have occurrences of threatened or endangered species.

Subwatersheds were then identified for projects based on:

- Willingness of landowner to participate;
- Biotic and abiotic conditions; and
- Location and size of project area.

With 15 of the 25 farms with secured funding being completed, the outcome of this project thus far has resulted in 71,348 feet of stream being fenced from cattle, allowing riparian buffers to establish along many sensitive headwater streams, and 53.4 acres of wetlands protected to create additional riparian areas

free of agricultural practices. More detailed results of this project can be found in the Monitoring section of this chapter.

Lakes and Ponds

Lakes and ponds are valuable ecosystems providing habitats for plant and animal species. Some of these waterbodies are used for recreational purposes, such as swimming, fishing, and boating, while others may be used for flood control and/or drinking water sources.

The Shenango River watershed has two large reservoirs: the Pymatuning Reservoir and the Shenango River Lake. These reservoirs provide a reliable source of drinking water for downstream water suppliers, serving a total population in excess of 250,000.

As a result of nutrient loading from a variety of sources, these large, shallow reservoirs have become very biologically active, negatively impacting recreational use and drinking water quality. The high organic content in the raw water used by the downstream public drinking water suppliers creates difficult and expensive challenges to them and, ultimately, to the consumers. Several times each year, the extremely high biological activity results in the production of chemical compounds known as Geosmin and methylisoborneol (MIB). Depending on seasonal weather conditions, the episodes can be severe and persist for several months. Public water suppliers find that this problem is often impossible to treat adequately, and it is extremely expensive. While the water is safe to drink, consumers find the water quality objectionable even though primary drinking water regulations are not compromised. The only treatment removal technique effectively used by the water suppliers is activated carbon, either in the powdered form or using granular activated carbon filters. Treatment costs to deal with this specific problem can range from \$500,000 to \$800,000 per year and still not adequately address the problem. It is recommended that further study be done on these reservoirs to determine the exact causes and sources of the taste and odor issues (Baumann, personal communication 2003).

In the late 18th century, settlers from Connecticut and New York began to move westward into the northwestern portion of Pennsylvania. Many began to farm the gently rolling hills and flat plains, but found that in certain areas, the land supported rich harvests of grain, while other areas contained swampy depressions. One of the largest wetland areas was known as Pymatuning swamp, derived from the Iroquois, meaning “the Crooked-mouthed Man’s Dwelling Place,” with “crooked-mouthed” referring to deceit. Although Native Americans used the swamp as a hunting ground, these early farmers found the Pymatuning swamp to be a threat. It swallowed stray cattle and provided habitat to dangerous animals like bears, mountain lions, snakes, and disease-carrying mosquitoes.



Pymatuning Reservoir

In 1868, the General Assembly of Pennsylvania studied the feasibility of draining the Pymatuning swamp. The plan was abandoned when it was determined that draining it would hurt industries downstream. However, severe flooding in the early 19th century led to the construction of a flood control dam in 1932, creating the 23 square-mile (17,088 acre) lake. The **Pymatuning Reservoir** is located in Crawford County, Pennsylvania and Ashtabula County, Ohio, and forms part of the Ohio-Pennsylvania border. The primary purpose of the Pymatuning Reservoir is to conserve the waters entering Pymatuning swamp and to regulate the flow of the Shenango and Beaver Rivers. The secondary purpose is to use the

reservoir to impound floodwater during periods of excessive runoff from the 158 square miles of drainage area above the dam. The reservoir is 16 miles long and 2.5 miles wide.

The Pymatuning Reservoir has two natural areas. Black Jack Swamp consists of 725 acres and provides habitat for unique natural communities. In the northern part of the lake is Clark Island, which has 161 acres of mature hardwood and white pine forest. Visitors are welcome to explore these undeveloped natural areas. Since the reservoir saddles the state lines of Ohio and Pennsylvania, each state offers a park in the name of Pymatuning. The creation of the dam opened countless water recreation opportunities. Today, Pymatuning State Park (PA) offers marinas, boat launching facilities, ball fields, playgrounds, picnic areas, campgrounds, hunting grounds, hiking trails, and environmental education programs.



Shenango River Lake

The **Shenango River Lake**, a six square-mile reservoir, is situated within the suburban Shenango Valley in Mercer County, Pennsylvania and Trumbull County, Ohio. The lake was formed by the construction of the Shenango River Dam. This dam is a concrete gravity dam with an uncontrolled center spillway. The project was authorized by the Flood Control Act of 1938, completed construction in 1965, and began full operation in February 1967. Shenango River Lake is part of the flood-control system operated by the USACE for the Beaver and Upper Ohio Rivers. The dam is located on the Shenango River almost one mile north of Sharpsville, Pennsylvania. It is about 33 miles above the Shenango River's confluence with the Mahoning River near New Castle, Pennsylvania, where the Beaver River is formed. Shenango River Lake hosts a recreation area and is a popular site for many types of activities, including fishing, boating, camping, swimming, picnicking, and hiking.

Lake Latonka is located in Jackson and Coolspring townships, Mercer County. This is a 230-acre private lake formed by a dam on Cool Spring Creek. A residential development has been built around this lake.

There are an estimated 75 additional lakes and ponds in the Upper Shenango River subwatershed; 241 in the Middle Shenango River subwatershed; 123 in the Lower Shenango River subwatershed; and 448 in the Neshannock Creek/Big Run subwatershed. Of special note are Crystal Lake and Dollar Lake. These are two of only eight glacial lakes in Pennsylvania.

Water Quality

According to Title 25 of the Pennsylvania Code Chapter 93 Water Quality Standards, water may not contain substances attributable to point or non-point source discharges in concentrations or amounts sufficient to be unfavorable or harmful to the designated water uses or to human, animal, plant, or aquatic life. In addition to other substances listed within or addressed by Chapter 93, specific substances to be controlled include, but are not limited to, floating materials, oil, grease, scum, and substances that produce color, tastes, odors, turbidity, or settle to form deposits (Pennsylvania Code 1997).

Cold Water Fishery (CWF) and Warm Water Fishery (WWF) designations are based primarily on temperature and dissolved oxygen levels in the surface waters. To be considered a CWF or WWF, the stream must maintain specific temperatures during specific times of the year and not fluctuate from that temperature more than 2°F over a one-hour period as a result of any effluent entering the stream. A CWF must also maintain a daily average dissolved oxygen (DO) level of 6.0 mg/l, whereas a WWF must

maintain a daily average DO level of 5.0 mg/l (Pennsylvania Code 1997). OhioEPA uses a similar definition of designations, with a slightly different name: Warm Water Habitat (WWH), Cold Water Habitat (CWH), Exceptional Warm Water Habitat (EWH), Modified Warm Water Habitat (MWH), and Limited Resource Water (LRW). These designations can be found in Table 3-3.

Table 3-3. Shenango River and Tributaries Water Use Protections
(Source: Pennsylvania Code 1997; OhioEPA 2002)

Waterway	County	State	Use Designation
Upper Shenango River Subwatershed			
Shenango River	Crawford, Mercer, Lawrence	PA	WWF
Linesville Creek	Crawford	PA	WWF
Bennett Run	Crawford	PA	WWF
Paden Creek	Crawford	PA	WWF
Gravel Run (OH)	Ashtabula	OH	WWH
Black Creek (OH)	Ashtabula	OH	WWH
McMichael Creek (OH)	Ashtabula	OH	WWH
Pymatuning Reservoir	Ashtabula	OH	WWH
Middle Shenango River Subwatershed			
Sugar Run	Crawford	PA	WWF
Wade Run	Ashtabula	OH	WWH
Little Shenango River	Mercer	PA	TSF/WWF
Morrison Run	Mercer	PA	Unknown
Crooked Creek	Mercer	PA	TSF/WWF
Bluff Run	Mercer	PA	TSF/WWF
McDowell Run	Mercer	PA	TSF/WWF
Randolph Run	Mercer	PA	TSF/WWF
Unger Run	Mercer	PA	TSF/WWF
Jackson Run	Mercer	PA	TSF/WWF
Patton Run	Mercer	PA	TSF/WWF
Mathay Run	Mercer	PA	WWF
Big Run	Mercer	PA	WWF
Lawango Run	Mercer	PA	WWF
Lackawannock Creek	Mercer	PA	TSF/WWF
Daley Run	Mercer	PA	WWF
Magargee Run	Mercer	PA	WWF
Golden Run	Mercer	PA	WWF
Brush Run	Mercer	PA	WWF
Pymatuning Creek	Mercer / Ashtabula, Trumbull	PA/OH	WWF/WWH
Berry Creek	Ashtabula	OH	WWH

Table 3-3. Shenango River and Tributaries Water Use Protections (continued)

Waterway	County	State	Use Designation
<i>Middle Shenango River Subwatershed (continued)</i>			
Chestnut Run	Mercer	PA	WWF
Clear Creek	Trumbull	OH	WWH
Maple Creek	Trumbull	OH	WWH
Stratton Creek	Trumbull	OH	WWH
Sugar Creek	Trumbull	OH	WWH
Mill Creek	Trumbull	OH	WWH
Booth Run	Mercer	PA	WWF
Pine Hollow Run	Mercer	PA	WWF
Saul Run	Mercer	PA	WWF
Shenango River Lake	Mercer	PA	WWF
<i>Lower Shenango River Subwatershed</i>			
McCullough Run	Mercer	PA	WWF
Thornton Run	Mercer	PA	WWF
Big Run			
Pine Run	Mercer / Trumbull	PA/OH	WWF/WWH
Yankee Creek	Trumbull	OH	WWH
South Branch	Trumbull	OH	WWH
Little Yankee Creek	Trumbull	OH	WWH
Deer Creek	Trumbull	OH	WWH
Bobby Run	Mercer	PA	WWF
Hogback Run	Mercer	PA	WWF
Turkey Run	Mercer	PA	WWF
Buchanan Run	Lawrence	PA	WWF
Deer Creek	Lawrence	PA	WWF
<i>Neshannock Creek/Big Run Subwatershed</i>			
Neshannock Creek	Mercer	PA	TSF/WWF
Otter Creek	Mercer	PA	TSF/WWF
Mill Run	Mercer	PA	TSF/WWF
Oilmill Run	Mercer	PA	TSF/WWF
Kent's Run	Mercer	PA	TSF/WWF
Munnell Run	Mercer	PA	TSF/WWF
Cool Spring Creek	Mercer	PA	TSF/WWF
Mill Creek	Mercer	PA	TSF/WWF
Yellow Creek	Mercer	PA	TSF/WWF
Fox Run	Mercer	PA	TSF/WWF
Beaver Run	Mercer	PA	TSF/WWF

Table 3-3. Shenango River and Tributaries Water Use Protections (continued)

Waterway	County	State	Use Designation
Neshannock Creek/Big Run Subwatershed (continued)			
Pine Run	Mercer	PA	TSF/WWF
Mill Run	Mercer	PA	TSF/WWF
Indian Run	Mercer	PA	TSF/WWF
Potter Run	Lawrence	PA	TSF/WWF
Little Neshannock	Mercer	PA	TSF/WWF
West Branch	Mercer	PA	TSF/WWF
Harthegig Run	Mercer	PA	TSF/WWF
Hottenbaugh Run	Lawrence	PA	TSF/WWF
Big Run	Lawrence	PA	WWF
WWF = Warm Water Fishery; CWF = Cold Water Fishery; TSF = Trout Stocked Fishery; WWH = Warm Water Habitat (Ohio designation)			

Point and Non-point Source Pollution

To effectively regulate and ultimately mitigate the mass load of pollutants entering streams, pollutant sources are classified into two main categories: point and non-point source pollution. Point source discharges are regulated under the National Pollutant Discharge Elimination System (NPDES) permit, established by Section 404 of the Clean Water Act of 1972. A listing of NPDES permits in the watershed can be found in Appendix G. Point source pollutants can be easily traced to their source, such as discharges from industrial or municipal facilities. Non-point source pollutants, sometimes called “runoff pollution,” typically have no readily visible source and often require detailed analysis and research to discern the source. Common sources of non-point pollution are abandoned mines, agriculture, urban runoff, construction activities, malfunctioning on-lot sewage systems, and forestry runoff.

PA DEP's plan for achieving comprehensive assessment of its surface waters includes the Statewide Surface Water Assessment Program (SSWAP) to evaluate all unassessed free-flowing streams using a field-level biological assessment. A major purpose of SSWAP is to delineate areas with water quality impairment and determine the type of pollution responsible. Information from this program, along with data from intensive surveys and other sources, is considered in the development of the Section 303(d) list of waters that may trigger the development of TMDLs, which are discussed in the ‘Pennsylvania’s Impaired Waters’ section of this chapter.

According to the 1999 Pennsylvania Non-point Source Management Plan, the Shenango River watershed has been identified as a high priority agricultural watershed for the Section 319 Nutrient Management Project. This project involved the employment of a nutrient management technician in the sponsoring county, which in this case was Crawford County, with Mercer County Conservation District also participating. Objectives of this project were to hire and train nutrient management personnel, develop nutrient management plans, and to complete best management practice demonstration projects.

Nearly one-third of the entire watershed is comprised of agricultural lands, including cropland, hay land, and pastures (Crawford County Conservation District et al. 1991). The 1990 Water Quality Assessment indicates that agriculture presents a threat to nearly all areas of the watershed, including the middle Shenango River, Pymatuning Creek, Neshannock Creek, Little Neshannock Creek, Otter Creek,

and Cool Spring Creek. The potential problems presented by the agriculturally dominant areas include biochemical oxygen demand, ammonia, fecal coliform, excess nitrification, erosion, and sedimentation.

Pennsylvania's Impaired Waters

The PA DEP has developed a program to assess the quality of waters in Pennsylvania and identify streams and other waterbodies that do not meet water quality standards. The goal is to protect uses the water can support, such as aquatic life, recreation, and potability (drinking water). There are numerical and/or narrative water quality criteria that express the in-stream levels of substances that must be achieved to support the uses. Periodic reports are required under section 305(b) of the federal Clean Water Act. Figures 3-11, 3-12, and 3-13 show impaired waters within the Middle Shenango River subwatershed, Lower Shenango River subwatershed, and Neshannock Creek/Big Run subwatershed.

Section 303(d) of the federal Clean Water Act requires Pennsylvania to identify all impaired waters within the state where technology-based treatment requirements for point and non-point sources of pollution are not stringent enough to attain and/or maintain applicable water quality standards. This list of impaired streams is called the Integrated Waterbody List, formerly known as the 303(d) and 305(b) lists. The Integrated Waterbody List includes water quality limited segments that require the development of total maximum daily loads (TMDLs) to assure future compliance with water quality standards (Table 3-4). Segments for which the source is unknown are listed as either "other" or "unknown."

A TMDL is a document that identifies allowable pollutant loads from point and non-point sources to a specific waterbody. A TMDL also includes a margin of safety to ensure protection of the water. Water quality limited segments are defined as waterbodies that do not meet water quality standards even after the application of technology-based treatment requirements to point and non-point sources of pollution. Water quality standards consist of three components, including: water uses to be protected, water quality criteria necessary to protect those uses, and an anti-degradation statement designed to protect existing water quality and uses.

Table 3-4. Impaired Streams and Lakes Requiring TMDLs

Stream Name	Code	Source	Cause	List Date	TMDL Date	Down RM*	Up RM*	Total Miles	Use Assessed
Bobby Run	35940	Other	Nutrients	1998	2007	0	3.59	3.6	Aquatic Life
Bobby Run (UNT 35942)	35942	Other	Nutrients	1998	2007	0	0.54	0.5	Aquatic Life
Cool Spring Creek (UNT 35820)	35820	Agriculture	Siltation	1998	2007	0.15	1.96	1.8	Aquatic Life
Crooked Creek	36175	Hydromodification	Siltation/Organic Enrichment/Low D.O.	1998	2007	0	2.14	2.1	Aquatic Life
Crooked Creek	36175	Hydromodification	Siltation/Organic Enrichment/Low D.O.	1998	2007	2.14	6.95	4.8	Aquatic Life
Crooked Creek	36175	Hydromodification	Siltation/Organic Enrichment/Low D.O.	1998	2007	6.95	7.93	1	Aquatic Life

Table 3-4. Impaired Streams and Lakes Requiring TMDLs (continued)

Stream Name	Code	Source	Cause	List Date	TMDL Date	Down RM*	Up RM*	Total Miles	Use Assessed
Crooked Creek	36175	Hydromodification	Organic Enrichment /Low D.O.	1998	2007	7.93	8.39	0.5	Aquatic Life
Deer Creek	35888	Unknown	Nutrients	2002	2007	0	3.05	3	Aquatic Life
Fox Run	35793	Abandoned Mine Drainage/Natural Sources	Metals/Organic Enrichment, Low D.O.	1998	2011	0.49	2.44	2	Aquatic Life
Little Shenango River (UNT 36286)	36286	Agriculture	Nutrients	1998	2007	0	1.83	1.8	Aquatic Life
Magargee Run (UNT 36056)	36056	Package Plants	Nutrients	2002	2007	0	1.08	1.1	Aquatic Life
Mill Creek	35754	Abandoned Mine Drainage	Metals	1998	2011	1.61	2.56	1	Aquatic Life
Mill Creek (UNT 35762)	35762	Abandoned Mine Drainage	Metals	1998	2011	0	0.22	0.2	Aquatic Life
Neshannock Creek (UNT 35516)	35516	Abandoned Mine Drainage/Urban Runoff	Metals/Storm Sewers, Siltation	1998	2011	0	1.58	1.6	Aquatic Life
Neshannock Creek (UNT 35533)	35533	Agriculture	Siltation	1998	2007	0	0.89	0.9	Aquatic Life
Neshannock Creek (UNT 35644)	35644	Agriculture	Siltation	1998	2007	0	1.63	1.6	Aquatic Life
Otter Creek	35679	Agriculture/Natural Sources	Nutrients/Organic Enrichment, Low D.O.	1998	2007	11.62	13.1	1.5	Aquatic Life
Otter Creek (UNT 35738)	35738	Agriculture/Natural Sources	Nutrients/Organic Enrichment, Low D.O.	1998	2007	0	1.25	1.2	Aquatic Life
Shenango River	35482	Other	Nutrients	1996	2007	2.99	15.6	12.6	Aquatic Life
Shenango River	35482	Other	Metals	2002	2007	2.99	15.6	12.6	Aquatic Life
Shenango River	35482	Agriculture	Pathogens	2004	2013	5.11	6.1	1	Human Health
Shenango River	35482	Other	Nutrients	1996	2007	15.57	24.2	8.6	Aquatic Life
Shenango River	35482	Other	Metals	2002	2007	15.57	24.2	8.6	Aquatic Life

Table 3-4. Impaired Streams and Lakes Requiring TMDLs (continued)

Stream Name	Code	Source	Cause	List Date	TMDL Date	Down RM*	Up RM*	Total Miles	Use Assessed
Shenango River	35482	Other	Nutrients	1996	2007	25.13	28.1	2.9	Aquatic Life
Shenango River	35482	Other	Metals	2002	2007	25.13	28.1	2.9	Aquatic Life
Shenango River	35482	Hydromodification	Nutrients	1996	2007	32.94	33.8	0.9	Aquatic Life
Shenango River	35482	Hydromodification	Organic Enrichment/Low D.O.	1998	2007	62	66.7	4.7	Aquatic Life
Shenango River (UNT 35850)	35850	Package Plants	Nutrients	2002	2007	0	2.62	2.6	Aquatic Life
Yellow Creek	35778	Abandoned Mine Drainage	Metals	1998	2011	0	7.83	7.8	Aquatic Life
Yellow Creek (UNT 35785)	35785	Abandoned Mine Drainage	Metals	1998	2011	0	1.21	1.2	Aquatic Life
Yellow Creek (UNT 35786)	35786	Abandoned Mine Drainage	Metals	1998	2011	0	0.58	0.6	Aquatic Life

Lake Name	Source	Cause	List Date	TMDL Date	Acres	Use Assessed
Lake Latonka	Unknown	Organic Enrichment/Low D.O.	2002	2007	320	Aquatic Life

*Down RM = Downstream River Mile; Up RM = Upstream River Mile

PA DEP must develop TMDLs for each waterbody on the 303(d) list. A TMDL is designed to reduce pollutant loads to impaired waters and enable those waters to meet water quality standards. Pennsylvania has committed to developing TMDLs for all impaired waterbodies and will use both traditional and new approaches to correct water quality problems.

Two TMDLs were completed and approved by EPA in 2001 for the Shenango River. Both TMDLs are for a 33.8-mile segment of the Shenango River. This segment was assessed through the surface water monitoring program, and the causes of impairment were PCBs and Chlordane, the sources of which were unknown. Table 3-5 shows a list of streams and lakes impaired by pollution that do not require a TMDL. These segments have been exempt from completing a TMDL because they are not impaired by a pollutant, but rather by an alternate source of impairment such as channelization or habitat modification.

PA DEP also lists stream segments that are expected to meet designated uses within a reasonable timeframe, and therefore do not require a TMDL to be established. There are two such stream segments on the Shenango River that have been assessed for aquatic life use. One is a 1.6-mile segment impaired by unionized ammonia (from RM 1.62 to 3.22), and the other is a two-mile segment (from the mouth to RM2.04) impaired by nutrients. The source of impairment for both segments is from package plants.

Table 3-5. Streams and Lakes Impaired by Pollution not Requiring a TMDL

Stream Name	Watershed	Code	Source	Cause	List Date	Down RM	Up RM	Total Miles	Use Assessed
Bobby Run	20A	35940	Other	Other Habitat Alterations	1998	0	3.59	3.6	Aquatic Life
Bobby Run (UNT 35942)	20A	35942	Other	Other Habitat Alterations	1998	0	0.54	0.5	Aquatic Life
Cool Spring Creek (UNT 35820)	20A	35820	Other	Water/Flow Variability	1998	0.15	1.96	1.8	Aquatic Life
Little Neshannock Creek (UNT 35542)	20A	35542	Channelization	Flow Alterations	1998	0	2.6	2.6	Aquatic Life
Little Neshannock Creek (UNT 35542)	20A	35542	Road Runoff	Water/Flow Variability	1998	0	2.6	2.6	Aquatic Life
Pine Run	20A	35951	Habitat Modification	Other Habitat Alterations	1998	0	1.72	1.7	Aquatic Life
Pine Run	20A	35951	Habitat Modification	Flow Alterations	1998	0	1.72	1.7	Aquatic Life
Pine Run (UNT 35952)	20A	35952	Habitat Modification	Other Habitat Alterations	1998	0	1.41	1.4	Aquatic Life
Shenango River (UNT 35924)	20A	35924	Other	Other Habitat Alterations	1998	0	2.04	2	Aquatic Life

Lake Name	Watershed	Source	Cause	List Date	Acres	Use Assessed
Pymatuning Reservoir	20A	Other	pH	2002	14,500	Aquatic Life
Shenango River Lake	20A	Other	pH	2002	3,041.30	Aquatic Life

There are also two TMDLs scheduled for completion in 2005. They are Cool Spring Creek, which is impaired by siltation from agriculture, and Neshannock Creek, also impaired by siltation from agriculture.

Waterbody assessment and data evaluation is a continuous process. The 2002 303(d) list was developed using information from stream and lake surveys and other sources, including PA DEP's Statewide Surface Water Assessment Program (formerly known as 'Unassessed Waters'), the Water Quality Assessment Process, the Non-point Source Program, and existing data submitted by external groups and agencies.

Table 3-6. Unassessed Streams and Lakes

Stream Name	Watershed	Stream Code	% Not Assessed	Total Stream Length (mi)
Little Deer Creek	20A	64901	100	1.59
Pymatuning Creek	20A	35985	100	5.32
Shenango River	20A	35482	37	33.9

Table 3-6. Unassessed Streams and Lakes (continued)

Lake Name	Watershed	Acres
All Unnamed Lakes	20A	46.71
Buhl Lake	20A	6.04
Dollar Lake	20A	0.49
Lower Lake	20A	8.90
Mud Lake	20A	16.98

Ohio's Impaired Waters

The Ohio 2004 Integrated Water Quality Monitoring and Assessment Report is similar to Pennsylvania's Impaired Waterbody List. The report indicates the general condition of Ohio's waters and identifies waters that are not meeting water quality goals. Like Pennsylvania, Ohio's report has combined the Section 305(b) water quality reports and the Section 303(d) list of impaired waters. Table 3-7 shows the impaired waters within the Shenango River watershed in Ohio. Figures 3-11, 3-12, and 3-13 show impaired waters within the Middle Shenango River subwatershed, Lower Shenango River subwatershed, and Neshannock Creek/Big Run subwatershed.

Table 3-7. Ohio EPA 2004 Integrated Report on Impaired Waterways

Assessment Unit	Code	Size (sq mi)	Impairment of Aquatic Life Use	Impairment of Recreation Use	Next Field Monitoring	Projected TMDL Date
Tributaries to Pymatuning Reservoir (within Ohio)	5030102010	40.7	Unknown	Unknown	2008	N/A
Pymatuning Creek	5030102030	149.5	Yes	Unknown	2008	2010
Yankee Creek; Little Yankee Creek	5030102050	94.6	Yes	Unknown	2008	2010

Stormwater Runoff

Stormwater runoff occurs as a result of impervious cover, which is material that covers the land that water cannot penetrate. Examples of impervious cover include roads, parking lots, buildings, sidewalks, and driveways. In a forested or vegetated area, rain soaks into the ground and is either taken up by tree or plant roots, moves down through the soil and into the groundwater, or runs off the land. When rain falls on impervious cover, it cannot soak into the ground and instead becomes stormwater runoff. When it rains on impervious cover, stormwater runoff is increased by 16 times, as compared to a vegetated or forested area.

The amount of impervious cover in a watershed influences dry and wet weather stream flow, channel shape and size, water quality, and plant and animal habitat. Most streams receive their base flow from groundwater. Impervious surfaces block water from contributing to groundwater, and lower stream flows can occur during dry weather. In addition, during rain events water runs off impervious surfaces, enters the storm drain system, and many times is directed straight to the stream. This large amount of stormwater runoff into the stream system can cause more frequent flooding, and higher level flooding. Increased flooding also results in accelerated streambank erosion, enlarged channel width, sedimentation on the stream bottom making the stream shallower, and straightening of the stream channel. When it rains, pollutants that have built up on impervious surfaces are washed into the stream. Examples of such

